

Spatially-optimized photovoltaic site selection in Algeria: Assessing solar potential using high-resolution data, GIS, and multicriteria analysis

Y. Halimi ^{a,*}, S. Halimi ^d, Z. Bouzid ^{b,a}, N. Ghellai ^{c,a}

^a Material and Renewable Energies Research Unit (URMER), University of Abou Bekr Belkaid, Tlemcen, Algeria

^b University of Ben Youssef Ben Khadda, Algiers, Algeria

^c Institute of Applied Sciences and Technologies, University of Abou Bekr Belkaid, Tlemcen, Algeria

^d MDN, Algiers, Algeria

ARTICLE INFO

Keywords:

Solar photovoltaic (PV) potential
Geographic Information System (GIS)
Multi-Criteria Decision-Making (MCDM)
Analytic Hierarchy Process (AHP)
Digital elevation model (DEM)
Algeria
Solar atlas
High-resolution suitability map

ABSTRACT

The identification of appropriate locations for photovoltaic (PV) solar power plants presents a multifaceted challenge that entails a complex interplay of diverse criteria. Algeria, which has a clear advantage in becoming a major player in the field of solar energy production, has in recent years intensified programs aimed at taking full advantage of this potential. Therefore, in order to provide the necessary assistance to the government and all the operators wishing to install photovoltaic systems in this country, this paper aims to present a research methodology that combines Geographic Information System (GIS) analysis, Multi-Criteria Decision-Making (MCDM) techniques, and off-site measurements to identify the most suitable sites for off-grid PV facilities installation across Algeria. However, before reaching the main objective, a refined digital elevation model (DEM) was developed. In addition, monthly mean values of global daily horizontal and tilted irradiance for a typical meteorological year were calculated, facilitating the mapping of these two quantities over the 12 months. Finally, the annual sum of photovoltaic energy produced by a PV power plant with a total installed capacity of 1 MWp was calculated, ranging from 1535 MWh/MWp to 2051 MWh/MWp per year depending on geographic location. The Analytic Hierarchy Process (AHP) was used to evaluate multiple criteria and prioritize the most suitable areas, classifying Algeria into seven distinct grades: “Somewhat suitable”, covering 36.4030% of the total country, followed by “Moderately suitable” at 32.4613% and “Very suitable” at 11.2877%. “Marginally suitable” lands comprise 15.8632%, while “Highly suitable” and “Minimally suitable” areas account for 2.0042% and 1.7342%, respectively. The “Extremely suitable” sites cover the smallest area at 0.2463% and are located primarily in the Southeast and East–Southeast regions. The outcome of this research is an updated Solar Atlas of Algeria and a high-resolution suitability map showing the spatio-temporal variability of PV potential across the country.

1. Introduction

Efficient energy usage is essential for countries to gain a competitive edge in the global market and achieve sustainable development. Economically prosperous nations are those that skillfully utilize energy resources, thereby enhancing productivity and minimizing operational expenses. However, the reliance on fossil fuels for energy production, which has been prevalent for a long time, has had significant implications for the environment. The combustion of fossil fuels releases greenhouse gases, such as carbon dioxide (CO₂), water vapor (H₂O), methane (CH₄), and ozone (O₃) into the atmosphere, leading to many detrimental effects including global warming (enhanced greenhouse effect). This elevated temperature disrupts various components of the climate system, altering weather patterns, sea levels, and other factors,

ultimately giving rise to climate change and its dangerous nature. As a solution, there is a growing need to shift towards cleaner and more sustainable sources of energy to mitigate the impact of human activities which are causing a disruption in the balance and leading to harmful consequences on the environment.

In recent times, a significant global energy predicament has arisen since 2021, which continues to persist today. This predicament stems from a critical scarcity of energy resources, which has been exacerbated by the resumption of global economic activities following the COVID-19 outbreak in 2020 [1,2]. The resulted disruption in energy supply has raised concerns about achieving sustainable development goals and transitioning to clean energy. In particular, numerous households have witnessed a remarkable surge in the adoption of photovoltaic

* Corresponding author.

E-mail addresses: yasmine.halimi@univ-tlemcen.dz (Y. Halimi), z.bouzid@univ-alger.dz (Z. Bouzid), nassghellai@gmail.com (N. Ghellai).

<https://doi.org/10.1016/j.enconman.2024.119176>

Received 19 July 2024; Received in revised form 17 October 2024; Accepted 18 October 2024

Available online 7 November 2024

0196-8904/© 2024 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

technology (which is one form of solar energy) amidst their financial struggles resulting from the prevailing energy crisis [3].

Solar energy comes from an inexhaustible source and is considered renewable, unlike fossil fuels. The abundant sunlight, in the form of electromagnetic radiation emitted by the Sun, enables the generation of solar energy, which is subsequently captured and converted into usable heat (solar thermal, solar thermodynamic) or electricity (solar photovoltaic). Sunlight is by far the largest carbon-free energy source on the planet. More energy from sunlight strikes the Earth in 1 h (4.3×10^{20} J) than all the energy consumed on the planet in a year (4.1×10^{20} J) [4].

To sustainably meet our daily energy needs and produce electricity cleanly, the deployment of photovoltaic (PV) installations becomes essential. Among the crucial steps involved in sizing a photovoltaic system, one integral part is the assessment of solar potential. It refers to the capacity of a region to harness solar energy, influenced by factors like location, climate, and sunlight availability. Studying solar potential allows for the determination of the minimum and maximum irradiance levels in a selected area. This information helps to optimize the design, configuration, and performance of the PV system and maximize energy production. It also aid in estimating financial returns and payback periods while ensuring system reliability.

Algeria, a country in North Africa, harbors the most considerable solar potential within the Mediterranean basin as stated by the German Aerospace Center (DLR) [5]. This finding underscores the growing recognition of the profitability and dependability of photovoltaic solar energy in regions characterized by abundant solar resources. Notably, Algeria emerges as a prominent exemplar within these areas, boasting the highest solar potential across the entire Mediterranean region. This country receives a total of 3500 h of sunshine annually, as determined by the Ministry of Energy and Mines [6]. The magnitude of solar resources is remarkably substantial to the extent that it could surpass the country's overall power production by a factor of 5000 [7]. To fully exploit this potential, the Algerian government is allocating significant resources to invest in renewable energy sources, with a specific focus on solar energy. Furthermore, the government is actively promoting and supporting research and development endeavors pertaining to renewable energy sources, while also offering tax credits and subsidies to companies engaged in initiatives aimed at solar energy production. In 2019, FARES et AL. conducted an extensive investigation on the exploitation of solar energy in Algeria. The study elucidated Algeria's potential in the realm of renewable energy sources and underscored the significance of political determination and economic management in optimizing their utilization [8].

The National Renewable Energy Development Program of Algeria, initiated in 2011, encompasses ambitious targets for diverse energy sources, including solar, wind, biomass, thermal, and geothermal. By the year 2030, Algeria aspires to attain a total production capacity of 12,000 megawatts for domestic consumption and 10,000 megawatts for export. Notably, the program has allocated a significant portion towards the achievement of 13,575 megawatts of photovoltaic energy, in addition to other energy sources [9]. This strategic positioning of Algeria renders it poised for substantial industrialization and competitiveness [10]. According to recent statistics, in the year 2022, the cumulative solar power capacity in Algeria surged to an impressive magnitude of 460 MW [11], whereas 4 years ago this country had 24 photovoltaic power stations with a total capacity of 344 MW [12]. This marks a remarkable achievement in the nation's renewable energy sector.

Significant research in Algeria has focused on solar potential, particularly in the field of photovoltaics. Studies primarily center around measuring, evaluating and forecasting various types of solar radiation, including total horizontal, diffuse, and direct normal irradiance. Additionally, research efforts also encompass the development of solar models and the validation of meteorological data measurements.

Solar irradiation models are crucial for designing and optimizing solar energy systems, such as photovoltaic systems. By simulating the solar radiation at a specific location, these models help determine the potential energy output, size, and orientation of solar panels. In 2006, A. MOUMMI et AL. undertook a research study wherein they conducted a comparative analysis of both the "Perrin de Brichambaut" and the "Liu and Jordan" models aimed at estimating solar radiation in Algeria, with a specific focus on the region of Biskra [13]. Results showed that the "Liu and Jordan" model closely matched experimental data during sunrise and sunset, while the "Perrin de Brichambaut" model was more accurate around solar noon. In 2007, S. BENKACIALI published a study wherein his team employed their equipment to comprehensively characterize the Ghardaïa site in terms of its solar potential [14]. The results showed the effectiveness and accuracy of installed radiometric station at the Ghardaïa Renewable Energy Research Unit. In 2016, a study by Z. BOUZID et AL. analyzed the complementarity of solar and wind energy in Algeria [15]. By evaluating temporal variations using the Pearson correlation coefficient, the researchers identified regions suitable for hybrid systems without storage. The following year, HASSAINE and MRAOUI utilized the PV Geographic Information System (PVGIS) and a solar radiation database to analyze solar energy resources and determine the photovoltaic potential in Algeria [16]. The influential parameters for PV production systems, including climatic and technological factors, were identified. The findings showcased a significant PV potential in the country, mainly in the Southern region. The study emphasized the need for further research, site assessments, and adaptation of existing technologies for successful photovoltaic technology implementation.

The solar field in Algeria has not only been extensively studied by many researchers but also in a multitude of diverse forms, encompassing solar energy potential evaluation [17], forecasting global solar irradiance for various resolutions using time series models [18], estimation of clear sky global solar radiation by evaluating three different models [19], and numerous other methodologies. Among these methods, the utilization of satellite Earth observation data, particularly MODIS products from NASA [20], is noteworthy. Additionally, the exploration of integrating Geographic Information Systems (GIS) with Multi-Criteria Decision Making (MCDM) techniques for optimizing the placement of photovoltaic solar power plants is significant. In 2021, SETTOU et AL. [21] presented a method for site selection of large-scale grid-connected solar PV systems in Algeria, combining GIS with the Analytical Hierarchy Process (AHP). The research tested different raster resolutions (92 m to 1000 m) to assess their impact on identifying suitable sites, revealing that a 92 m resolution could designate 17% of the area as optimal for generating over 136 million GWh of energy. However, the study exhibits certain limitations regarding the accuracy of the digital elevation model (DEM) in comparison to the data provided by NASA. Additionally, the solar irradiation maps lack precision, as they are based on only 216 data points, leading to discrepancies in solar irradiation estimations across different regions. In the same year, HADDAD et AL. developed a methodology for mapping site suitability for concentrated solar power (CSP) plants in Algeria using MCDM and GIS [22]. They found that over 51% of the territory is unsuitable for CSP due to factors like topography and water availability, identifying optimal areas mainly in Naâma, Laghouat, and Ghardaïa. In the same field, GOUAREH et AL. [23] developed a GIS-based framework for site selection of large-scale concentrating solar power plants in Algeria, integrating MCDM techniques like AHP and BWM (Best Worst Method). Another study by DÍAZ-CUEVAS et AL. [24] on a GIS- and MCDM-based assessment of Algeria's renewable energy potential, focusing on wind and photovoltaic solar farms, indicated that 97% of Algeria is unsuitable for wind and PV plants; however, significant areas (25,377 km² for wind, 13,485 km² for solar, and 11,199 km² for both) show high potential. Their model emphasized the importance of extending the electric grid to enhance suitability.

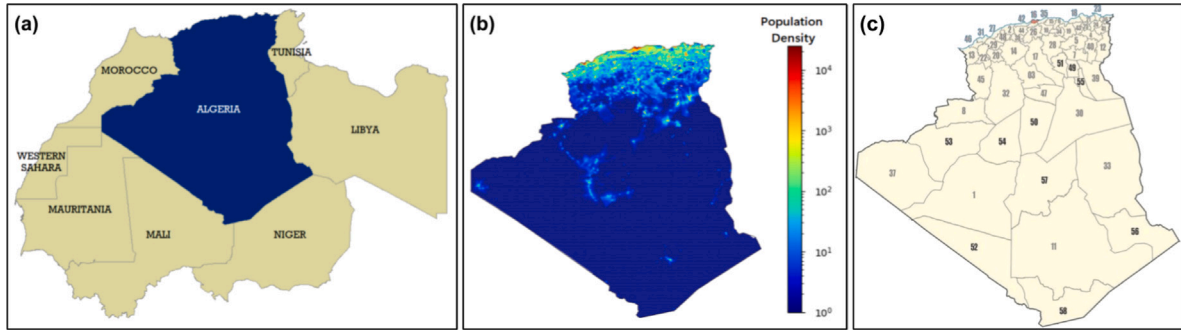


Fig. 1. Study area. (a) Geographical position of Algeria in North Africa. (b) Population density map of Algeria. (c) 58 wilayas (provinces) of Algeria according to the new distribution of 2019.

When seeking a summary of the solar energy potential at either a local or global scale, it is common to consult a Solar Atlas. For Algeria, the first Solar Atlas, created by CAPDEROU in 1986 [25], was a significant achievement in the field of solar energy. However, subsequent developments and improved measurement accuracy have revealed divergences between the atlas's results and reality [26]. As a result, new models have been proposed to enhance irradiation estimation. Lately, the CDER ("Centre de Développement des Energies Renouvelables" in french), which is a research center devoted to Development of Renewable Energies in Algeria, published and approved a new Algerian Solar Atlas. It provides solar resource evaluation in Algeria [27] by using measured time insolation to estimate global solar irradiation for different skies. The Atlas includes solar maps for various inclinations and orientations, benefiting scientists, socioeconomic stakeholders, and bioclimatic building design. In 2023, CDER published another Atlas that evaluated the solar resource and renewable energy potential in Algeria. This Atlas included estimates and maps of solar irradiation, energy production for 1 MWp solar power plants, and avoided carbon dioxide emissions for 64 sites across the Algerian national territory [28].

Through the conduct of a comprehensive analysis of the extensive body of research pertaining to the solar potential in Algeria, it becomes evident that the majority of these studies exhibit certain limitations and weaknesses. Some notable limitations include the presence of inaccurate and incomplete map data for the digital elevation model of Algeria, the unavailability and lack of verification regarding long-term classifications of climate maps within Algeria (such as those provided by the Köppen–Geiger classification system [29,30]), the limited data on solar radiation and its covered regions, and the absence of sufficient data within the country raster map to construct a comprehensive depiction of Algeria, which further hampers the pixel disaggregation of satellite images.

This research focusses on the spatio-temporal distribution of solar energy potential in Algeria, aiming to detect the most suitable sites in the country for the implementation of stand-alone PV systems. To achieve this, an optimized high resolution spatio-temporal solar photovoltaic potential map of Algeria is presented. Furthermore, this study seeks precise and up-to-date data for renewable energy planning, agriculture, urban development, conservation, and disaster management. The methodology integrates GIS analysis, MCDM techniques, and off-site measurements to comprehensively evaluate key factors. The analysis indicates that photovoltaic energy has the most significant impact on the implementation sites, followed by aspect, slope, sunshine duration, and population density as the least influential factor.

The main novelties of this study are:

- The analysis encompasses the entire geographical area of Algeria, with more than 4.180589×10^{10} recorded values of daily incident global insolation on a horizontal surface spanning from 01/01/1984, to 31/12/2021.
- By applying solid mathematical methodologies to the obtained data, optimized global horizontal and tilted irradiation maps, as well as the corresponding optimal tilt angles, were generated.

- The effect of temperature on solar PV efficiency was considered in order to generate map illustrating the photovoltaic potential in Algeria.
- To strengthen the site selection process, a refined DEM for Algeria has been developed. This enhanced model extends the captured elevation range from the original -51 to 2866 meters (NASA) to a new range of -48 to 2906 meters, offering greater precision and validation.
- To address the inaccuracies present in the climate zone maps of Algeria published by various sources, a revised map has been created utilizing the Köppen–Geiger classification, accompanied by an amended legend.
- By leveraging machine learning algorithms, along with easily obtainable parameters, a reliable model for predicting sunshine duration was developed and validated.
- The AHP was used to create a suitability map for Algeria, classifying the area into seven grades for photovoltaic system installation based on five criteria.

The comprehensive approach presented in this paper can serve as a model for similar assessments in other countries or regions seeking to harness the potential of solar energy resources. The layout of this article is as follows: the introduction provides context and literature review, while the methodology mainly details the GIS, solar radiation, influential and constraint factors, and multi-criteria analyses employed. The results present optimized climate, irradiation, and sunshine maps, and assess Algeria's solar PV viability. Finally, the conclusion summarizes the key technical insights on enhancing solar energy utilization in Algeria.

2. Methodology

2.1. Study area

Algeria, one of the member states of the Arab Maghreb Union (AMU), is a country located in North Africa, bordering the Mediterranean Sea (Fig. 1(a)). This country has a population of 44,758,398 inhabitants and covers an area of $2,381,740 \text{ km}^2$ [31]. It has held the distinction of being the largest country in both Africa and the Arab world since the separation of Sudan in 2011 [32]. This vast territory is divided into two parts by the Saharan Atlas: the north, which includes almost the entire population, and the Sahara to the south, which represents nearly 85% of the entire country's area [33,34].

Benefiting from its geographical position and considerable space, Algeria is a favorable land for the implementation of photovoltaic systems with an estimated solar potential of 1700 kWh/m^2 per year in the north and 2263 kWh/m^2 per year in the south [35]. The Fig. 1(b) depicts the variation in population density across the country, with the capital Algiers exhibiting the highest concentration of inhabitants. The data used to generate this figure was obtained from [36], with the most recent metadata available (updated in 2020) providing the necessary information.

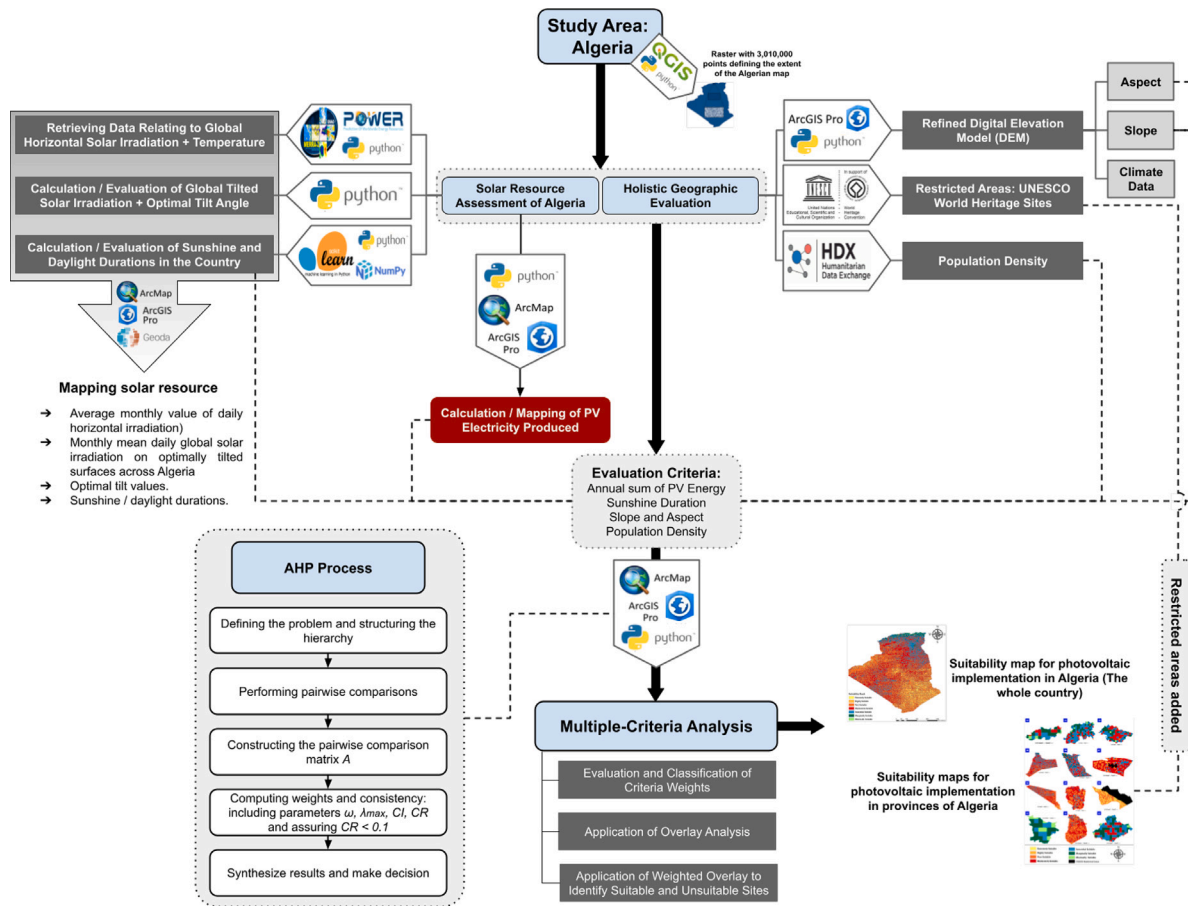


Fig. 2. Integrated methodology for determining the solar photovoltaic potential in Algeria.

2.2. Flowchart of methodology

In the field of renewable energy, publications often focus on the solar potential of a well-defined zone or specific regions within a country. Through geographical maps, the spatio-temporal distribution of the amount of solar radiation reaching the ground is represented. As mentioned in the introduction, the importance of this type of information in decision-making when it comes to choosing the most appropriate locations for the installation of solar energy systems is no longer to be demonstrated. However, when it comes to photovoltaic generators in particular, knowing the amount of solar radiation that can be collected as a function of the latitude ϕ and longitude ψ of the location as well as the time is not enough, because for this type of system, a larger number of functional parameters can influence the conversion efficiency and the PV electricity production.

Fig. 2 illustrates the flowchart of the methodological approach, which includes the key steps followed and described in this article. The presence of criteria considered influential and conditional for the placement of photovoltaic solar systems is noted. It is important to recognize that these factors may differ depending on the configuration of the photovoltaic system to be installed: either an autonomous system or a grid-connected system (in the latter case, preferred locations are those where the general electricity distribution network is already present). Based on the collected data, the focus is on the former type. Additionally, two multi-criteria decision support methods are utilized: Analytic Hierarchy Process (AHP) and Weighted Linear Combination (WLC). These methods serve to identify and rank the most suitable regions for potential autonomous ground-mounted photovoltaic installations.

2.3. Material and methods

2.3.1. The Geographical Information System (GIS)

A Geographical Information System (GIS) is a system of computer software, hardware and data, personnel that make it possible to enter, manipulate, analyze, and present data, and the information that is tied to a location on earth's surface [37,38]. The data used in this paper comprise two types: externally originated data obtained from reputable and established sources; and internally generated data derived through rigorous, well-defined mathematical models.

Quantum Geographic Information System (QGIS) is a free, open-source, cross-platform and scalable geographical information system tool with plugin development in Python and C++ languages [39]. This software enables users to generate, modify, view, scrutinize, and release geospatial data. It is intended to be compatible with a diverse array of vector and raster data formats, including but not limited to ESRI Shapefile and GeoTIFF. In this research, the version 3.28.2 of QGIS and 3.10 of Python were used to obtain the geographic coordinates of a raster with 3,010,000 points defining the extent of the Algerian map (Fig. 3). The average distance between each point and its nearest neighbors in the map varies across different directions, ranging from 300 to 700 m in the east–west direction, 1 to 1.8 km in the northeast–northwest direction, and 900 m to 1.5 km in the southeast and southwest directions.

ArcGIS, a system for people who rely on accurate geographic information to make decisions, is a suite consisting of a group of GIS software products produced by Environmental Systems Research Institute (ESRI) [41]. ArcMap and ArcGIS Pro, two specific software

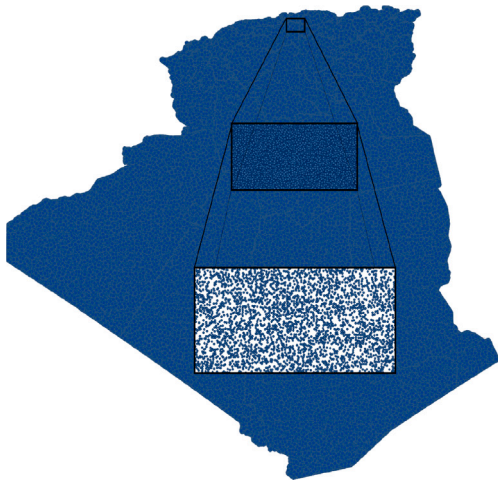


Fig. 3. Geographic coordinates of a raster with 3,010,000 points defining the extent of the Algerian map.

Table 1
The three letters of the Köppen–Geiger climate classes classification [40].

1st	2nd	3rd
A	f (Rainforest)	
	m (Monsoon)	
	w (Savannah)	
B	W (Desert)	h (Hot)
	S (Steppe)	k (Cold)
C	w (Dry winter)	a (Hot summer)
	f (No dry season)	b (Warm summer)
	s (Dry summer)	c (Cold summer)
D	w (Dry winter)	a (Hot summer)
	f (No dry season)	b (Warm summer)
	s (Dry summer)	c (Cold summer)
		d (Very cold winter)
E		T (Tundra)
		F (Frost)

products within the ArcGIS suite, were utilized to digitize the pertinent data for generating solar maps that depict the geographic distribution of the photovoltaic potential that can be recovered in Algeria.

2.3.2. Data collection and processing

2.3.2.1. Climate maps. Climate maps are useful resources that can illustrate the spatial patterns of various climatic variables, such as temperature, precipitation, wind direction and speed, atmospheric pressure, humidity, solar radiation, and cloud cover, among others. However, to better understand the patterns and characteristics of the world's climates, the Köppen–Geiger climate classification can be referenced. This widely used system categorizes climates based on their temperature and precipitation characteristics. It was originally developed by German climatologists Wladimir Köppen and Rudolf Geiger in the mid-20th century [29,30], and has undergone revisions and updates by other researchers over time. The Köppen–Geiger climate types are divided into five main groups: tropical (A), arid (B), temperate (C), continental or cold (D), polar (E), and further subcategorized based on additional factors such as seasonal variations, temperature extremes, and moisture availability (Table 1).

Algeria's unique geographic location results in a wide variety of climatic regions throughout the country. The Ministry of Housing provided two maps that represent the country different climate zones for both summer and winter [43]. These maps were readjusted by several authors in their research [42,44]. For winter, three major climatic zones were determined based on the influence of altitude

(Fig. 4(d)). During the summer period, five climatic zones are distinguished (Fig. 4(c)) [42]. In particular, the altitude legend provided in the last cited figures does not correspond to the exact data of Algeria, which may cast doubt on the veracity of the presented information. This matter is taken very seriously, and measures have been implemented to rectify it using a refined digital elevation model. In a subsequent section, a corrected altitude map will be provided along with a revised legend that accurately represents the data pertaining to the winter period.

Concerning the Köppen–Geiger classification, several climate maps are available based on information from the literature, including the one presented by [29] for the 50 year period 1951–2000 in a low resolution or a pixelated and aggregated form (particularly when zoomed in) (Fig. 4(a)). A better resolution for the case study is provided by [40] which presents a high resolution for the Algerian climate map over a 36 year period 1980–2016 (Fig. 4(b)). In this research paper, with the aim of offering information that may not be apparent in shorter-term datasets, climate data for Algeria were sourced from the WorldClim database [45], encompassing minimum and maximum temperatures and precipitation over a 62-year period (1960–2021) at a spatial resolution of 2.5 min (approximately 21 km at the equator). This resulted in 120 high-definition GeoTIFF files per month, totaling 2232 files for the three parameters, with each file approximately 6 GB in size. Utilizing the extent shapefile of the country, the global climate data were clipped to focus solely on the Algerian territory. The Typical Meteorological Year (TMY) was computed for both precipitation and average temperature across the 62-year dataset, which reduced the file size significantly. Then, data cleansing was performed to address missing values (NaN) arising from scanned pixels in the raster. Subsequently, the files were resampled to adjust pixel size, ensuring compatibility with a specific grid without losing critical information. Finally, 3,010,000 data points were categorized into corresponding climate zones using the Köppen–Geiger classification, facilitating the generation of detailed climate maps as presented in the results.

2.3.2.2. Solar radiation data and temperatures. The surface observation network of the Algerian National Meteorology Office (ONM in french which stands for “Office National de la Météorologie”) currently consists of 80 meteorological stations distributed across the country (with a higher density in Northern regions), while the altitude network comprises 5 stations [46]. These stations continuously measure a set of parameters related to weather forecasts, including solar radiation received at the Earth's surface. Unfortunately, access to these data is not freely available and is difficult to obtain. Additionally, they are limited by:

- The number of regions they cover (especially in the Southern regions);
- The number of years they span;
- The format and temporal frequency at which they are provided.

The research focuses on retrievable photovoltaic potential, necessitating data related to solar radiation. To address the challenges presented by the ONM, reliance has been placed on the NASA Earth Science Research Program, which supports satellite systems that provide significant research data for the study and analysis of climatic processes. A web portal, known as POWER (Prediction Of Worldwide Energy Resources), has been established for accessing this data.

The POWER solar data is based upon satellite observations from which surface insolation values are inferred. The meteorological parameters are based upon the MERRA-2 assimilation model [47,48]. Using Python, a modern programming language, an application was developed to access the POWER data repository. With input of 3,010,000 latitude ϕ and longitude ψ coordinates, the application automatically retrieved the values of daily incident global insolation on a horizontal surface H in kWh/m² day for each (ϕ, ψ) pair and for each day of the year, starting from 01/01/1984 until 31/12/2021. For the entire territory

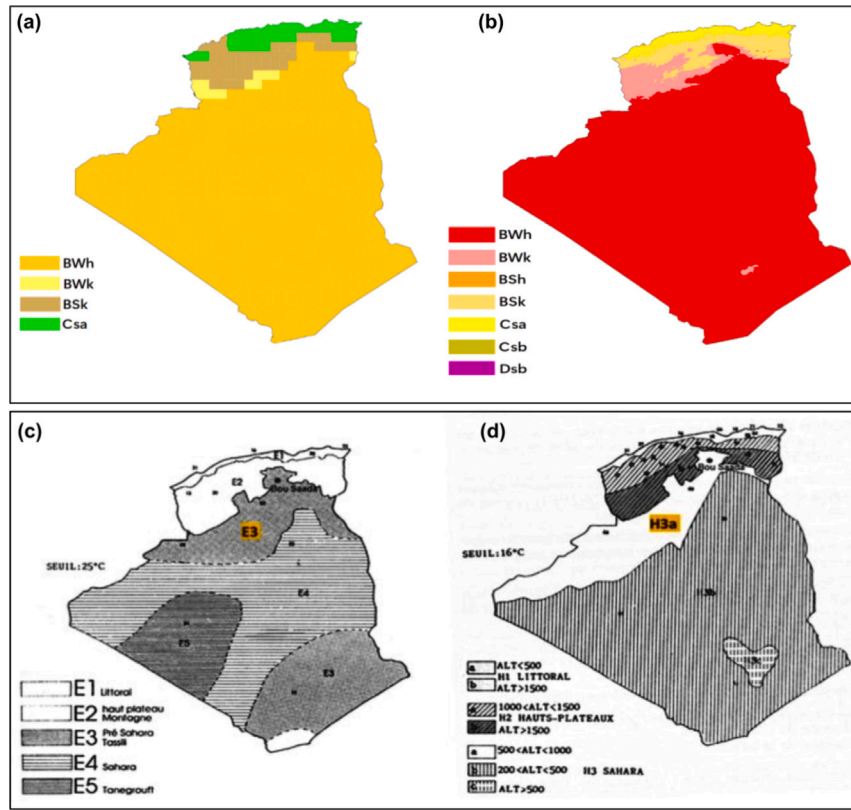


Fig. 4. (a) Climatic map of Algeria according to the Köppen–Geiger classification for the period 1951–2000 (by [29]). (b) Climatic map of Algeria according to the Köppen–Geiger classification for the period 1980–2016 (by [40]). (c) Climatic zoning in Algeria for the summer period (adapted by the author from the work of [42]). (d) Climatic zoning in Algeria for the winter period (adapted by the author from the work of [42]).

Table 2
Algerian sites inscribed on the UNESCO World Heritage list (WHS).

Properties	Year of inscription	Type	Document reference
Al Qal'a of Beni Hammad	1980	Cultural	[49]
Djémila	1982	Cultural	[50]
M'zab Valley	1982	Cultural	[50]
Timgad	1982	Cultural	[50]
Tipasa	1982	Cultural	[50]
Tassili n'Ajjer	1982	Mixed (cultural and natural)	[50]
Kasbah of Algiers	1992	Cultural	[51]



Fig. 5. Location of Algerian sites inscribed on the UNESCO World Heritage list (WHS).

of Algeria, approximately 4.180589×10^{10} values of H were obtained and stored in CSV files (each file for a locality). Subsequently, these data were processed to obtain, for each month of the TMY corresponding to a location in the country, an average monthly value of daily horizontal radiation $\bar{H}$ (meaning that for each ϕ and ψ , a typical averages are calculated corresponding to the 12 months of the year). By exploiting these data, it was possible to plot the global horizontal irradiation maps of Algeria, which are commented on and discussed in the results section.

Concerning the temperature parameter T , the same methodology was employed as for retrieving the H values, utilizing the POWER service. These T values represent the ambient temperature at approximately 2 m above the ground level.

2.3.2.3. Cultural and natural heritage sites. Approved in 1972, the “convention concerning the protection of the World’s Cultural and Natural Heritage” was adopted by the United Nations Educational, Scientific and Cultural Organization (UNESCO). The purpose of this convention is to “ensure the identification, protection, conservation, presentation

and transmission to future generations of cultural and natural heritage of outstanding universal value” [52].

Algeria has 7 properties inscribed on the UNESCO World Heritage List (WHS) (Table 2 and Fig. 5): six of cultural nature and one of mixed nature (cultural and natural). These properties can be considered as constraints with respect to the installation of photovoltaic infrastructures. Additionally, six sites are inscribed on the tentative list, a type of inventory of properties that each country proposes for inscription on the WHS [53].

2.3.2.4. Digital elevation model and derivatives. A Digital Elevation Model (DEM) is defined as an ordered or unordered digital set of

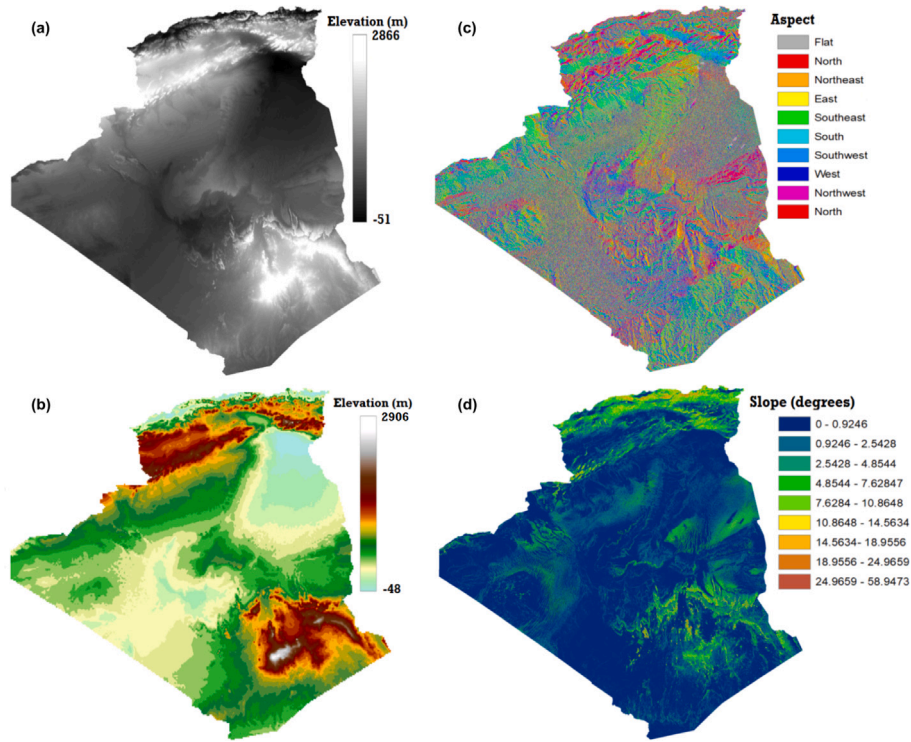


Fig. 6. (a) DEM of Algeria (data from NASA [55]). (b) Refined digital elevation model (created by the authors). (c) Aspect map for Algeria. (d) Slope map for Algeria.

ground elevation (spot height) for terrain representation, which is supposedly “bare” land surface model [54]. In this research, an enhanced, refined DEM was developed derived from the provided DEM data by NASA [55] for the study area encompassing the entire extent of Algeria.

To obtain this refined DEM with an expanded range of values (Fig. 6(b)), an approach that involves both disaggregation and aggregation steps was applied. Starting with the foundational DEM (Fig. 6(a)), computational processing and advanced analysis techniques were applied using both GIS tools and the programming language Python to extract more detailed spatial information from the original dataset. This disaggregation step resulted in expanding the DEM to over 49 million data points, providing a higher-resolution and more granular representation of the terrain. After that, the dataset was pixelized. By applying this aggregation step, the higher level of detail captured in the disaggregated DEM was smoothed out, resulting in a more manageable dataset. The increased level of detail and accuracy was maintained while reducing computational complexity.

Through this analysis, the elevation range captured in the refined DEM was expanded compared to the original DEM, with values spanning from −48 to 2906 meters, as opposed to the original DEM’s range of −51 to 2866 meters. This indicates that the enhanced model contains a more granular and higher-resolution representation of the terrain, providing greater detail and accuracy.

Using (Fig. 6(b)) in conjunction with GIS tools, the following information was derived:

- **Aspect:** it is the compass direction that a topographic slope faces, usually measured in degrees from north. Aspect can be generated from continuous elevation surfaces [56]. Considering the geographical location of Algeria in the northern hemisphere, it is recommended to accord precedence to the allocation of land oriented towards the south for the purpose of establishing ground-based photovoltaic power plants. Aspect map for Algeria is represented in Fig. 6(c).
- **Slope:** when considering the installation of a ground-based photovoltaic system, the inclination of the terrain assumes significant importance. Optimal conditions are attained on flat land with a

slope of 0°. Conversely, the prospect of employing an inclined terrain with a slope surpassing the critical threshold (equivalent to the latitude of the location) is strongly discouraged for photovoltaic installations. Therefore, the refined DEM serves to facilitate an analysis of the slope within the appointed study area (Fig. 6(d)).

2.3.3. Photovoltaic electricity generation and availability

The quantity and availability of photovoltaic energy are criteria considered as influential and conditional factors for the placement of PV systems. Thus, the following discussion focuses on the methods that enable the acquisition of such information.

2.3.3.1. Average irradiation on sloped surfaces. For use in the photovoltaic field, where the panels are often installed at an angle inclined relative to the horizontal (with β being the tilt angle), monthly average of the daily global irradiation on the inclined surface must be known.

If $\bar{H}$ represents the average daily global radiation on a horizontal surface (for a given month), its equivalent diffuse $\bar{H}_d$ can be calculated using the ERBS et AL. correlations [57]:

- If $\bar{\omega}_s \leq 81.4^\circ$ and $0.3 \leq \bar{K}_T \leq 0.8$:

$$\frac{\bar{H}_d}{\bar{H}} = 1.391 - 3.560\bar{K}_T + 4.189\bar{K}_T^2 - 2.137\bar{K}_T^3 \quad (1)$$

- If $\bar{\omega}_s > 81.4^\circ$ and $0.3 \leq \bar{K}_T \leq 0.8$:

$$\frac{\bar{H}_d}{\bar{H}} = 1.311 - 3.022\bar{K}_T + 3.427\bar{K}_T^2 - 1.821\bar{K}_T^3 \quad (2)$$

$\bar{\omega}_s$ represents the average monthly value of the Sun’s hourly angle at sunset (in the following, the formula for a single day is provided), and $\bar{K}_T$ represents the average monthly clearness index:

$$\bar{\omega}_s = \cos^{-1} [-\tan(\phi) \tan(\delta)] \quad (3)$$

$$\bar{K}_T = \frac{\bar{H}}{\bar{H}_0} \quad (4)$$

$\overline{H}_0$ represents the monthly average of the daily extraterrestrial radiation. The value of the solar declination in radians for the n th day of the year (n_{day}) is [58]:

$$\delta = 0.006918 - 0.399912 \cos(B) + 0.070257 \sin(B) - 0.006758 \cos(2B) + 0.000907 \sin(2B) - 0.002697 \cos(3B) + 0.00148 \sin(3B) \quad (5)$$

with:

$$B = (n_{day} - 1) \frac{360}{365} \quad (6)$$

By using the daily values for ω_s , δ , and the solar constant G_{sc} equal to 1367 W/m^2 [59], the daily extraterrestrial radiation can be calculated using the following equation [58]:

$$H_0 = \frac{24G_{sc}}{\pi} \left[1 + 0.033 \cos\left(\frac{360n_{day}}{365}\right) \right] \times \left[\cos(\phi) \cos(\delta) \sin(\omega_s) + \frac{\pi\omega_s}{180} \sin(\phi) \sin(\delta) \right] \quad (7)$$

The value of H_0 in Eq. (7) is given in Wh/m^2 day. The quantity $\overline{H}_T$, monthly mean daily global solar radiation on an inclined surface, can be calculated using the method of LIU and JORDAN [58,60]:

$$\overline{R} = \frac{\overline{H}_T}{\overline{H}} = \left(1 - \frac{\overline{H}_d}{\overline{H}} \right) \overline{R}_b + \frac{\overline{H}_d}{\overline{H}} \left[\frac{1 + \cos(\beta)}{2} \right] + \rho_g \left[\frac{1 - \cos(\beta)}{2} \right] \quad (8)$$

The value of the ground albedo, denoted as ρ_g , can vary depending on the nature of the ground surface. In this work, an average value of 0.2 has been selected for the entire Algerian territory.

Photovoltaic panels must be oriented towards the southern cardinal direction if they are installed in the northern hemisphere (which is the case for Algeria). So, for the average value of the parameter $\overline{R}_b$, it is calculated as follows [58]:

$$\overline{R}_b = \frac{r_1 + r_2}{r_3 + r_4} \quad (9)$$

with:

$$r_1 = \cos(\phi - \beta) \cos(\overline{\delta}) \sin(\overline{\omega}_s) \quad (10)$$

$$r_2 = \frac{\pi}{180} \overline{\omega}_s \sin(\phi - \beta) \sin(\overline{\delta}) \quad (11)$$

$$r_3 = \cos(\phi) \cos(\overline{\delta}) \sin(\overline{\omega}_s) \quad (12)$$

$$r_4 = \frac{\pi}{180} \overline{\omega}_s \sin(\phi) \sin(\overline{\delta}) \quad (13)$$

$\overline{\omega}_s$ is the monthly average value of the solar hour angle at sunset for an inclined surface. In the northern hemisphere, it is calculated using the following formula [58,61]:

$$\overline{\omega}_s = \min \left[\begin{array}{l} \cos^{-1} \left[-\tan(\phi) \tan(\overline{\delta}) \right] \\ \cos^{-1} \left[-\tan(\phi - \beta) \tan(\overline{\delta}) \right] \end{array} \right] \quad (14)$$

Several studies have found that during the summer period, the incident solar insolation is maximized for a surface inclined at about 15° below the value of the latitude. In the winter, an additional 15° above the latitude is taken [58,62–65]. In this study, inclined surface angles from 0° to 90° in steps of 1° were considered to identify the angles that maximize monthly mean daily global solar irradiation.

Using the results of all the calculations performed, maps of monthly average daily global solar irradiation on inclined surface, as well as those of the optimal inclination values, were plotted. These maps are commented and discussed on in the results section. Also, using the Python SciPy library [66,67], a linear regression was performed to estimate the relationship between the optimal tilt angle β_{opt} and the

latitude ϕ of the studied location. This relationship takes the following form:

$$\beta_{opt} = a_\beta \phi + b_\beta \quad (15)$$

with the values of a_β and b_β calculated and grouped in the Table 9.

2.3.3.2. PV electricity production formula. A photovoltaic solar system converts light into electricity. However, before this electricity is available to the consumer, the conversion process goes through several phases and experiences various losses due to different causes. To determine the amount of electricity generated by a total photovoltaic surface $S_{pv,total}$ for a given day n_{day} of the year where the daily global solar radiation on an inclined surface H_T has been calculated, the following equation may be utilized:

$$E_{pv} = S_{pv,total} \times H_T \times K \times \eta_{pv,ref} \quad (16)$$

It should be noted that E_{pv} is the daily energy generated per tilted area from the photovoltaic panels; $\eta_{pv,ref}$ is the rated conversion efficiency of photovoltaic modules; and a TMY containing the values of H_T , by exploiting the formulas (1) through (14), has been created by using daily values instead of considering the average monthly values.

The performance ratio of the PV system K is calculated by [68]:

$$K = K_{HD} \times K_{PD} \times K_{PM} \times K_{PA} \times K_{PT} \quad (17)$$

where K_{HD} is the coefficient that takes into account losses on the photovoltaic array surface; K_{PD} takes into account time-dependent losses by the module function; K_{PM} for losses by the array load; K_{PA} for losses by the array circuit; and K_{PT} is the module temperature coefficient (relation (18)). Just as outlined by [68], the following values were utilized: $K_{HD} = 0.97$, $K_{PD} = 0.95$, $K_{PM} = 0.94$, $K_{PA} = 0.97$. For K_{PT} the values were calculated using following equation:

$$K_{PT} = 1 + \alpha_{pv,max} \times \frac{T_C - T_{ref}}{100} \quad (18)$$

where $\alpha_{pv,max}$ represents the temperature coefficient for the maximum power output of the module, $T_{ref} = 25^\circ \text{C}$ is the reference temperature, as photovoltaic panels are typically tested in the industry under Standard Test Conditions (STC) of 25°C with an irradiance of 1000 W/m^2 , and T_C is the temperature of the module, which can be calculated as follows [68]:

$$T_C = T_A - \Delta T \quad (19)$$

With T_A being the ambient temperature; $\Delta T = 18.4^\circ \text{C}$ for photovoltaic systems mounted on racks [68].

2.3.3.3. Sunshine duration. The sunshine duration is derived from measurements of direct normal solar radiation (DNI). According to the standards of the World Meteorological Organization (WMO), it is defined as the duration during which the DNI reaches or exceeds the threshold of 120 W/m^2 [69]. In the absence of the required measurements of DNI , the Support Vector Machine (SVM) [70], along with the Linear Regression Method (LRM) [71] are utilized as machine learning algorithms, to estimate the sunshine duration. This approach is inspired by an influential study [72]. The aforementioned article proposed the utilization of machine learning algorithms and easily obtainable parameters to estimate sunshine duration in Saudi Arabia. These parameters included maximum possible daylength S_0 (Eq. (20)), extraterrestrial solar radiation H_0 , latitude ϕ , longitude ψ , altitude z , and month number.

$$S_0 = \frac{2}{15} \cos^{-1} [-\tan(\phi) \tan(\delta)] \quad (20)$$

Support vector machine offers exceptional capabilities in data analysis and prediction. In this study, the Python programming language was utilized to implement SVM. In the latter, SVR machine learning algorithm was chosen for regression tasks (SVR stand for ‘‘Support Vector Regression’’, a type of SVM). This involves the use of modules

Table 3

Some relevant studies using GIS and AHP for identification of the most suitable sites for solar energy exploitation.

Country	Year	Approach	Reference
Algeria	2021	GIS-BWM-AHP	[23]
	2021	GIS-AHP	[21]
	2021	GIS-AHP	[22]
	2021	GIS-AHP	[24]
Tunisia	2021	GIS-AHP	[74]
	2023	GIS-AHP	[75]
	2023	GIS-Fuzzy AHP (FAHP)	[76]
Morocco	2021	GIS-AHP	[77]
	2022	GIS-AHP	[78]
Egypt	2022	GIS-AHP	[79]
Brazil	2020	GIS-AHP-TOPSIS	[80]
Saudi Arabia	2018	GIS-AHP	[81]
	2021	GIS-AHP	[82]
ECOWAS region	2018	GIS-AHP	[83]
Turkey	2019	GIS-AHP	[84]
	2020	GIS-AHP	[85]
	2021	GIS-AHP	[86]
Spain	2020	GIS-PROMETHEE-AHP	[87]
	2024	GIS-AHP	[88]
Peru	2021	GIS-AHP	[89]
Iran	2022	GIS-Fuzzy-Boolean logic-AHP	[90]
Indonesia	2020	GIS-AHP	[91]

such as NumPy [67] and scikit-learn [73]. First, the dataset was pre-processed by extracting the relevant features, including S_0 , H_0 , ϕ , ψ and z . Additionally, the parameter day number n_{day} was incorporated, instead of month number, to achieve a more detailed understanding of the subtle variations in sunshine duration. However, it is important to note that the daily calculations entail certain implications where it significantly increases the computational time and expanding the size of the dataset. Instead of working with 12 months of data for each latitude and longitude, the analysis must handle 365 days of data for each location and its elevation.

After preprocessing, the model was trained using the prepared dataset (daily values for each feature). During this phase, SVR effectively acquired knowledge of the underlying patterns and relationships between the input features and the corresponding sunshine duration values. By optimizing the model parameters and employing appropriate kernel functions, the model's ability to accurately comprehend the variations and complexities within the daily data was confirmed. Consequently, a scaling factor of 0.8 was obtained, signifying the connection between daylight hours and extraterrestrial irradiation. This factor enables conclusive predictions regarding the sunshine duration for the specified coordinates.

Concerning the linear regression method, which is a statistical algorithm that establishes a linear relationship between an independent variable and a dependent variable, NumPy and scikit-learn were also utilized, followed a similar process to the SVR approach for collecting and preparing the dataset. In particular, only the data inputs and their formatting were different as they are specific to the regression task and focus on finding the best linear relation between the features and the target variable (the sunshine duration hours in this case), rather than separating data points into different classes. Once the dataset was preprocessed and the necessary parameters were extracted, the next step involved training the LRM model. During this phase, the latter learned the coefficients and intercept that define the linear relationship between the independent variables and the target variable. It is noteworthy to mention that LRM assumes a linear relationship between the features and the target variable. If the relationship is not linear, the model may not accurately capture the underlying patterns and variations in the data.

Table 4

AHP solution process [99].

Step 1	Identifying the problem
Step 2	Identifying criteria/alternatives
Step 3	Creating the hierarchical structure
Step 4	Normalization of binary comparison matrix
Step 5	Calculation of priority vectors
Step 6	Performing consistency tests
Step 7	Creating the decision matrix
Step 8	Calculation of final priority vectors

For the implementation of these two methods, in addition to Python, GIS tool and Geoda software was utilized for data interpolation calculations and mapping purposes. Geoda is a dedicated software tool primarily designed for exploratory spatial data analysis (ESDA). It specializes in reading Shapefiles, the common geospatial vector data format used in GIS software like ArcGIS [92]. After collecting the predictions of sunshine duration using both methods, in the event of missing results for variables at unobserved locations within a geographical area, spatial interpolation methods were applied such as Kriging and spline interpolation to address this issue. These techniques are utilized to estimate the missing values and ensure a more complete and accurate representation of the data on geographical heatmaps.

2.3.4. Multiple criteria methods for photovoltaic implementation sites

Multi-Criteria Decision Making (MCDM) refers to a family of decision-making methods that involves structuring and solving decision and planning problems that have multiple, often conflicting, criteria that need to be considered simultaneously. Thus, it is important to acknowledge that any decision made may suboptimal, and can potentially be improved upon when additional relevant information becomes available. Conversely, a complete absence of information will inherently preclude the application of any logical decision-making framework [93].

MCDM methods include several techniques developed by different authors such as ELECTRE, ZIONTS-WALLENUS interactive method, decision support system, nonlinear programming, ANP (Analytic Network Process), and AHP (Analytical Hierarchy Process). These two last methods were developed in the 1970s by THOMAS L. SAATY, and are both used to measure intangibles using human judgement [94].

2.3.4.1. Analytical Hierarchy Process theory. The Analytical Hierarchy Process (AHP) is a preeminent multi-criteria decision making methodology in comparison to alternative techniques. It is flexible, intuitive, and checks for inconsistencies. The hierarchical structure of AHP clarifies the importance of each element, enabling unbiased decision-making. While AHP has limitations like ranking irregularities, its advantages, such as requiring fewer pairwise comparisons, make it a more efficient approach than alternatives like the ANP [93]. Within the MCDM framework, analytical hierarchy process technique has been extensively employed in numerous studies [95–98], and in recent years, the GIS-AHP approach has been widely used to identify the most suitable sites for solar energy projects (Table 3).

The general procedural steps to be considered in the AHP method as explained by SAATY are tabulated in Table 4. Furthermore, the main pillars of AHP can be resumed in seven points as explained by [100]:

- (1) Ratio scales, proportionality, and normalized ratio scales.
- (2) Reciprocal paired comparisons.
- (3) Sensitivity of the principal right eigenvector.
- (4) Homogeneity and clustering.
- (5) Synthesis that can be extended to dependence and feedback.
- (6) Rank preservation and reversal.
- (7) Group judgments.

Table 5
The SAATY rating scale [94].

Intensity of importance	Definition	Explanation
1	Equal importance	Two factors contribute equally to the objective
3	Somewhat more important	Experience and judgement slightly favor one over the other
5	Much more important	Experience and judgement strongly favor one over the other
7	Very much more important	Experience and judgement very strongly favor one over the other. Its importance is demonstrated in practice
9	Absolutely more important	The evidence favoring one over the other is of the highest possible validity
2,4,6,8	Intermediate values	When compromise is needed

Table 6
Decision matrix of 5 criteria based on 10 pairwise comparisons.

	Photovoltaic Energy	Aspect	Slope	Sunshine duration	Population density
Photovoltaic Energy	1.00	3.00	5.00	7.00	9.00
Aspect	0.33	1.00	3.00	5.00	7.00
Slope	0.20	0.33	1.00	3.00	5.00
Sunshine duration	0.14	0.20	0.33	1.00	3.00
Population	0.11	0.14	0.20	0.33	1.00

The AHP scale, a tool used to determine the relative importance or priority of different criteria or alternatives, is commonly used in decision-making and multi-criteria analysis. This scale is a 9-point scale that ranges from 1 (equal importance) to 9 (extreme importance) as observed from the Table 5.

In the analytical hierarchy process theory, [94] has provided and demonstrated the use of mathematical formulas to achieve the complete procedure of the AHP method. Considering n elements $C_1, \dots, C_n$ with associated relative weights or priorities a_{ij} , forming a reciprocal matrix $A = (a_{ij})$ where $a_{ij} = \frac{1}{a_{ji}}$ and $a_{ii} = 1$. If the weights exhibit transitivity, such that $a_{ik} = a_{ij}a_{jk}$ for all i, j , and k , then the matrix A is consistent (such a matrix might exist if the a_{ij} are calculated from exactly measured data). For a consistent matrix, there exists an eigenvector ω of order n such that $A\omega = \lambda\omega$, where $\lambda = n$.

However, human judgements are often inconsistent, so the condition $a_{ik} = a_{ij}a_{jk}$ may not hold. In this case, the principal eigenvalue λ_{max} of A satisfies $\lambda_{max} \geq n$, and the difference $\lambda_{max} - n$ indicates the degree of inconsistency. The consistency index is calculated as $\frac{(\lambda_{max} - n)}{(n-1)}$. This index can be compared to the consistency index of random matrices of the same order, yielding a consistency ratio. SAATY suggested an acceptance value of 0.1 for the consistency ratio; however, higher values may be acceptable in certain circumstances. A consistency ratio of 0 indicates perfect consistency. This procedure can be resumed in the following five steps:

1. Multiply each column of the pairwise comparison matrix by the corresponding weight:

$$A.C = \begin{bmatrix} a_{11} & \dots & a_{1m} \\ a_{21} & \dots & a_{2m} \\ \dots & \dots & \dots \\ a_{m1} & \dots & a_{mm} \end{bmatrix} \begin{bmatrix} C_1 \\ C_2 \\ \dots \\ C_m \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \\ \dots \\ x_m \end{bmatrix} \quad (21)$$

2. Calculating the eigenvector λ by the following equation:

$$\begin{bmatrix} \lambda_1 \\ \lambda_2 \\ \lambda_3 \\ \dots \\ \lambda_n \end{bmatrix} = \begin{bmatrix} \frac{x_1}{c_1} \\ \frac{x_2}{c_2} \\ \frac{x_3}{c_3} \\ \dots \\ \frac{x_n}{c_n} \end{bmatrix} \quad (22)$$

3. Computing the average of the values, denote it by λ_{avg} which is maximum eigenvalue of the pairwise comparison matrix:

$$\lambda_{avg} = \frac{\|\sum \lambda_i\|}{\|n\|} \quad (23)$$

4. Calculating consistency index CI :

$$CI = \frac{\|\lambda_{avg} - n\|}{\|n - 1\|} \quad (24)$$

5. Calculating final consistency ratio CR :

$$CR = \frac{CI}{RI} \quad (25)$$

If this ratio $\frac{CI}{RI}$ (RI being random index) is very large (SAATY suggests > 0.1), it indicates insufficient consistency, and the best course of action is to revisit and revise the comparisons.

2.3.4.2. Weighted Linear Combination theory. The utilization of GIS in Multi-Criteria Evaluation (MCE) is primarily achieved through two distinct methodological approaches. The first method involves the employment of Boolean overlay, whereby all evaluative criteria are reduced to logical statements of suitability, which are subsequently combined through the application of one or more logical operators. Conversely, the second approach, known as Weighted Linear Combination (WLC), entails the standardization of continuous criteria to a common numeric range, followed by their integration through the calculation of a weighted average [101].

The WLC method is one of the most widely used GIS-based decision rules. It is often applied in land use/suitability analysis, site selection, and resource evaluation problems [102]. This method involves the following general steps:

- Identification of the relevant decision criteria (factors) and their relative importance (weights).
- Standardization of the factor values to a common scale, typically 0 to 1.
- Multiplication of the standardized factor values by their corresponding weights
- Summation of the weighted factor values to produce an overall suitability or decision score.

In the context of geographic information systems, the WLC method is commonly used for overlaying selected reclassified criteria. Reclassification refers to the process of transforming the original data values within a GIS dataset into new, more meaningful values that better align with the analysis objectives. The overlaying process is a common operation that involves combining multiple spatial data layers to create a new layer or derive new information [103]. This step is crucial as it allows the input criteria to be standardized and made comparable before being combined using the WLC method. By applying weights to the reclassified criteria, the weighted linear combination method allows to reflect the relative importance of each factor in the final output (a single suitability or decision map), thus providing a more informed decision-making process. The suitability is described by [104]:

$$S = \sum w_i x_i \quad (26)$$

where S is the suitability, w_i is the weight of factor i , and x_i is the criterion score of factor i . In cases where Boolean constraints apply, the

Table 7
Summary of weighted raster layers with their grades for climate, topography, and demographic criteria.

Criteria	Criteria weight	Factor weight	Factor	Classes	Grades
Climate-related	0.48	0.36	Epv (Wh/m ² year)	1,984,252,800 – 2,051,221,888	9
				1,947,525,632 – 1,984,252,800	8
				1,915,411,072 – 1,947,525,632	7
				1,886,874,496 – 1,915,411,072	6
				1,845,741,056 – 1,886,874,496	5
				1,784,854,528 – 1,845,741,056	4
				1,706,868,480 – 1,784,854,528	3
				1,617,637,248 – 1,706,868,480	2
				1,535,674,880 – 1,617,637,248	1
		0.12	Sunshine duration (h)	9.893704 – 9.959392	9
				9.873043 – 9.893704	8
				9.854326 – 9.873043	7
				9.838011 – 9.854326	6
				9.824086 – 9.838011	5
				9.811269 – 9.824086	4
				9.798418 – 9.811269	3
				9.777026 – 9.798418	2
				9.738915 – 9.777026	1
Topography	0.44	0.24	Aspect (°)	157.668869 – 201.9487 South	9
				113.344009 – 157.668869 Southeast	7
				68.868332 – 113.344009 East	5
				27.420149 – 68.868332 Northeast	3
				–1 – 27.420149 North	1
				287.987335 – 325.520782 Northeast	3
				245.248459 – 287.987335 West	5
				201.9487 – 245.248459 Southwest	7
				325.520782 – 359.857117 North	1
		0.20	Slope (°)	0 – 0.663956	9
				0.663956 – 1.56098	8
				1.56098 – 2.886651	7
				2.886651 – 4.692309	6
				4.692309 – 7.124732	5
				7.124732 – 10.26394	4
				10.26394 – 14.228551	3
				14.228551 – 19.928308	2
				19.928308 – 32.704903	1
Demography	0.08	0.08	Population density (people/km ²)	8083.45752 – 13015.957031	9
				5293.026367 – 8083.45752	8
				3523.62207 – 5293.026367	7
				2062.797119 – 3523.62207	6
				1142.557251 – 2062.797119	5
				586.387268 – 1142.557251	4
				231.357361 – 586.387268	3
				55.409916 – 231.357361	2
				0.0123 – 55.409916	1

procedure should be modified by multiplying the suitability calculated from the factors by the product of the constraints [104]:

$$S = \sum w_i x_i \prod c_j \quad (27)$$

where c_j is the criterion score of the constraint j .

2.3.4.3. Data preprocessing step. In addition to the annual sum of the photovoltaic energy, four additional parameters have been selected for inclusion in the multi-criteria decision making methodology: sunshine duration, slope, aspect, and population density. The process of obtaining the five factors to be used as criteria in MCDM required extensive preprocessing to all of these selected parameters.

The data cleaning process commenced with a thorough examination of all data to remove duplicate values and address missing values. Subsequently, the data was transformed into a suitable format and combined into a unified dataset. Additionally, a final representative subset of the data was selected for analysis, which included the plotting map. Finally, the data was exported to a tabular format, ensuring that the values were consistently ordered by latitude and longitude across all files. Some raster values were mixed, necessitating lengthy and complex calculations to achieve the desired order.

Concerning the slope and aspect raster layers, they were generated using the refined DEM layer, as mentioned earlier in Fig. 6(b).

Both analytical methods AHP and WLC have been combined and used in this research study. The integrated approach key steps are

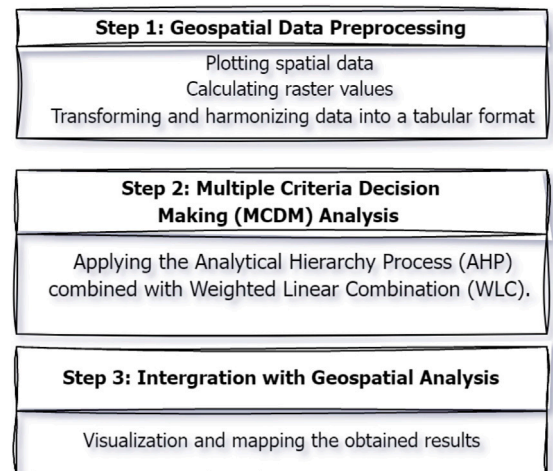


Fig. 7. Integrated approach for MCDM.

Table 8
Comparative table of elevation data from Halimi et al. NASA, and associated References.

N°	Name	Coordinates	Elevation in meters		
			Reference	HALIMI et AL.	NASA
1	Tahat	23.2864, 5.5338	2908 [105]	2902	2866
2	In Akoulmou	25.0833, 5.4166	2359 [106]	2013	1955
3	Chéla	35.3183, 6.6375	2328 [105]	2280	2236
4	Lalla Khedidja	36.4475, 4.2283	2308 [105]	2230	2218
5	Djebel Aïssa	32.9116, -0.4676	2236 [107]	2190	2176
6	Azao	25.2034, 8.1226	2158 [108]	2088	2005
7	Mount Babor	36.5500, 5.4666	2004 [109]	1180	1216
8	Mountain Isaak	36.5114, 5.0594	1742 [110]	1275	1174
9	Sidi Amar	35.8666, 1.6166	1985 [111]	1895	1193
10	Bouzegza	36.6065, 3.4505	1055 [112]	1000	940
11	Pic des Singes	36.7693, 5.0956	430 [113]	355	355
12	Djabel Ksel	34.0500, 2.0666	2008 [114]	1700	1484
13	Djabel Dirrah	36.2666, 2.7500	1485 [115]	1000	923
14	Chott Melhir	34.3333, 6.3333	- 40 [116]	-31	-30
15	Zaccar	36.1977, 2.2133	1554 [117]	280	283
16	Mansoura	36.1833, 4.4333	1862 [118]	553	553
17	Lac Oubeira	36.8447, 8.3894	23 [119]	22	22
18	Chot Hodna	35.3000, 4.6666	400 [120]	405	413
19	Chott Tinsilt	35.8833, 6.4833	792 [121]	791	784

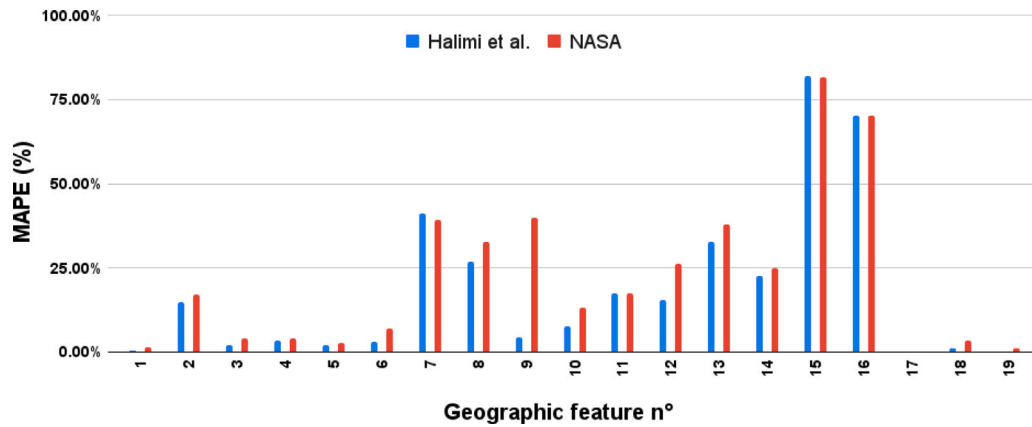


Fig. 8. Comparison of MAPE percentages for NASA and Halimi et al. elevation data.

resumed in Fig. 7. Based on the five criteria targeted for evaluation, a list of 10 unique pairwise comparisons has been generated. The decision matrix is tabulated in Table 6. Based on the performed comparisons of this matrix and using power numerical method, the principal eigenvalue λ_{max} is calculated to be approximately 5.3163. The consistency index of 0.0791 has been resulted. The random consistency index value is 1.12. Thus, the calculated consistency ratio gave a value of 7.06%. This value is less than the acceptable threshold of 10%. This indicates that the pairwise comparisons in the matrix have a reasonable level of consistency and can be considered reliable for the decision-making process.

Table 7 resumes the totality of obtained weights with their criteria and their classes of reclassified raster layers. For each data layer in this table, a reclassification scheme was developed that scores the suitability for photovoltaic (PV) implementation on a common scale from 1 to 9. This scoring is based on factors that influence PV performance and feasibility and has been applied to each factor layer.

2.3.5. Evaluation metric

To evaluate and validate the results, particularly in assessing the accuracy of the data obtained for the Algerian solar potential, the mean absolute percentage error (MAPE) is utilized. It is one of the most

widely used measures of forecast accuracy, due to its advantages of scale-independency and interpretability [122]:

$$MAPE = \frac{1}{N} \sum_{i=1}^N \left| \frac{X_i - Y_i}{X_i} \right| \quad (28)$$

X_i and Y_i denote the actual and forecast values at data point i , N is the number of data points.

3. Results and discussion

3.1. Analysis and validation of the refined DEM

Table 8 serves to validate the elevation results from the refined DEM. As demonstrated in the table, elevation measurements for 19 prominent geographic features were recorded, including lakes, mountains, and plateaus located in different regions in Algeria. These measurements are compared against data obtained from NASA's DEM and other relevant sources (data references).

For the majority of locations, the elevations recorded using the refined DEM were closer to the references values than those reported by NASA. This provides evidence that the results represented in the "HALIMI et AL." column, yield more accurate approximations of true

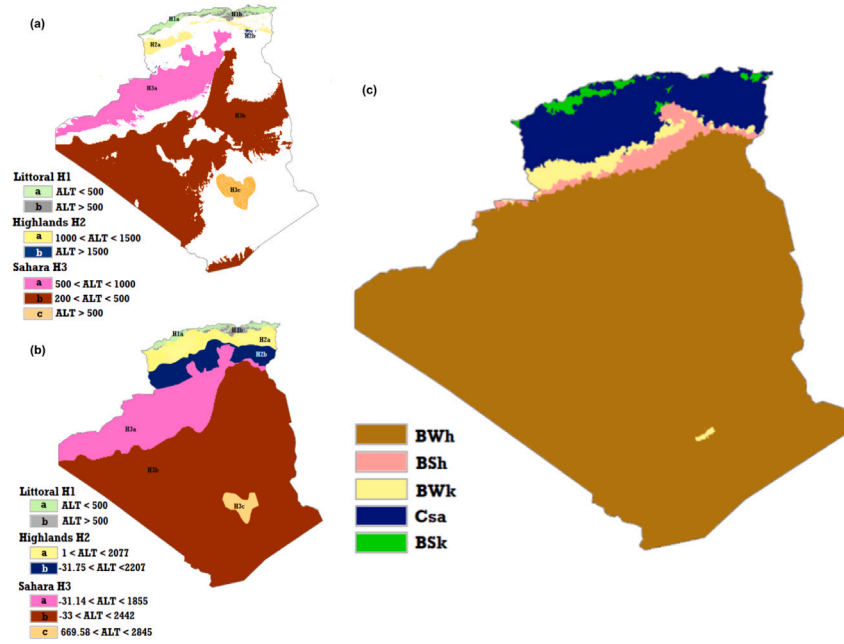


Fig. 9. (a) Actual altitudes compared to reported altitudes in the Algerian Ministry of Housing's winter climate zone map. (b) Revised winter climate zone map with accurate altitudes. (c) Climate zones of Algeria according to the Köppen–Geiger classification (updated by the Authors).

elevation compared to the alternative NASA dataset. The mean absolute percentage error was also calculated for each geographic feature (Fig. 8). Concerning the overall *MAPE* values, they are 22.42% for the NASA DEM and 18.38% for the refined DEM.

3.2. Correction and optimization of climate maps

In light of the preceding discussion regarding the climate zone maps provided by the Algerian Ministry of Housing, a significant discrepancy was discovered in the accuracy of the altitude legend accompanying these maps. Upon conducting a meticulous analysis, utilizing the refined digital elevation model that considered all altitude variation throughout the entirety of Algeria (Fig. 6(b)), it was observed that the altitudes depicted in the Ministry's map for the winter period (Fig. 4(d)) did not align with the actual altitudes (Fig. 9(a)). Indeed, several disparities have been identified and are hereby presented:

- One aspect of concern pertains to the highland region. The accompanying legend incorrectly designates this area as encompassing both H2a and H2b types in their entirety. However, upon meticulous examination, it becomes evident that the assigned two-colored zones do not accurately correspond to the actual distribution of altitudes within the highlands. A more precise analysis reveals that a smaller portion of the highland region exhibits altitudes aligning with the H2a and H2b classifications, consequently rendering the legend's representation erroneous.
- Similarly, the Sahara and littoral climate zones exhibits similar inconsistencies.

As it is imperative to rectify this discrepancy by accurately representing the altitudes corresponding to each climate zone entity, a newly created map has been carefully developed (Fig. 9(b)). This map incorporates the revised legend that precisely correspond to the altitudinal characteristics of each zone, ensuring a faithful cartographic illustration of the surface area within these regions.

Regarding the climate maps based on the Köppen–Geiger classification employed in this research, a low-resolution climate map spanning the 50-year period of 1951–2000 was initially referenced (Fig. 4(a)). This map contains only four climate types: BWh, BWk, BSk, and Csa. However, the limitations of this low-resolution map were recognized, as

it failed to provide the level of detail and accuracy required, especially when extracting pixel-level data or zooming in on smaller spatial scales. To address this issue, a higher-resolution alternative was actively sought, resulting in the acquisition of the map shown in Fig. 4(b). After analyzing this higher-resolution map, it is worth noting that it exhibited certain discrepancies or inaccuracies. Although the legend of the map indicated the presence of seven distinct climate types, only five (BWh, BWk, BSh, BSk, and Csa) were visibly depicted on the map itself. To address this discrepancy, a multi-decadal climate map was generated covering a period of 62 years, utilizing the Köppen–Geiger classification method (Fig. 9(c)).

Fig. 9(c) revealing the presence of five distinct climate types (BWh, BWk, BSh, BSk, and Csa) in the studied region. The absence of the Csb and Dsb climate types in the map suggests that these particular climate zones may not be as prevalent or significant in the area under investigation. In particular, the presence of the same climate types in both the resulted map and the Köppen–Geiger classification system substantiates the consistency and reliability of the findings. Nevertheless, it is noteworthy that some variations in the surface extent of each climate zone were observed when comparing the two maps (Figs. 4(b) and 9(c)). These observed differences suggest that certain climate zones may possess slightly larger or smaller spatial coverage than previously reported.

3.3. Global horizontal and tilted irradiation maps

3.3.1. Updating and optimizing maps

The analysis of the amount of average monthly value of daily horizontal solar irradiation $\bar{H}$ received on a horizontal surface, through the 12 maps plotted and represented in Fig. 10, shows that:

- Across the entire Algerian territory, this quantity tends to increase from January until it reaches its maximum in July (7.9721 kWh/m² day). From August, the received radiation begins to decrease until the end of the year when it reaches its minimum threshold.
- During the months of January, February, March, September, October, November, and December, the values of $\bar{H}$ in the entire Southern region of Algeria are higher compared to those in the

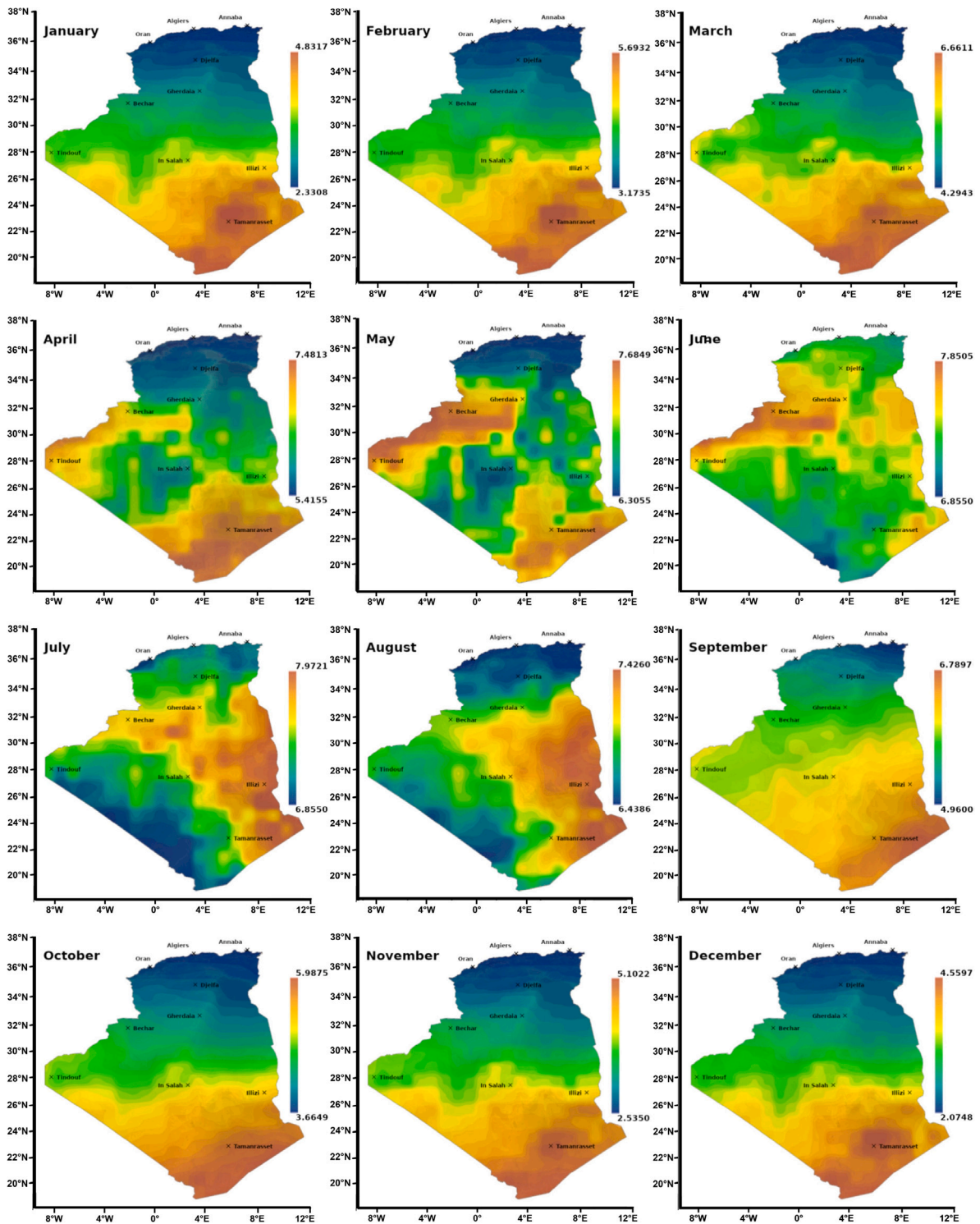


Fig. 10. Global horizontal irradiation maps of Algeria (average monthly value of daily horizontal irradiation). Legend values unit are in kWh/m² day.

North of the country. However, for the months of April, May, and June, it is noticed that the solar potential is more important in a part of the Southwest of Algeria (around Béchar and Tindouf) and in the extreme South (around Tamanrasset). During the months of July and August, the solar radiation reaches its maximum values throughout the Southeast Quadrant of Algeria.

- If one remains in the Northern region of the country throughout the entire year, it can generally be observed that the solar resource is weakest in the extreme Eastern part. This observation

can be supported at the coastal level by taking the capital, Algiers, as a reference point (36.7538 °N, 3.0588 °E, which is situated approximately in the middle). A comparison of the solar resources of all the sites to the West of Algiers with those to the east reveals that the solar resources to the east are significantly higher.

The analysis of the mean daily global tilted solar irradiation maps (Fig. 11) following the optimal tilt angle β (Fig. 12), for the 12 months of the year, shows that:

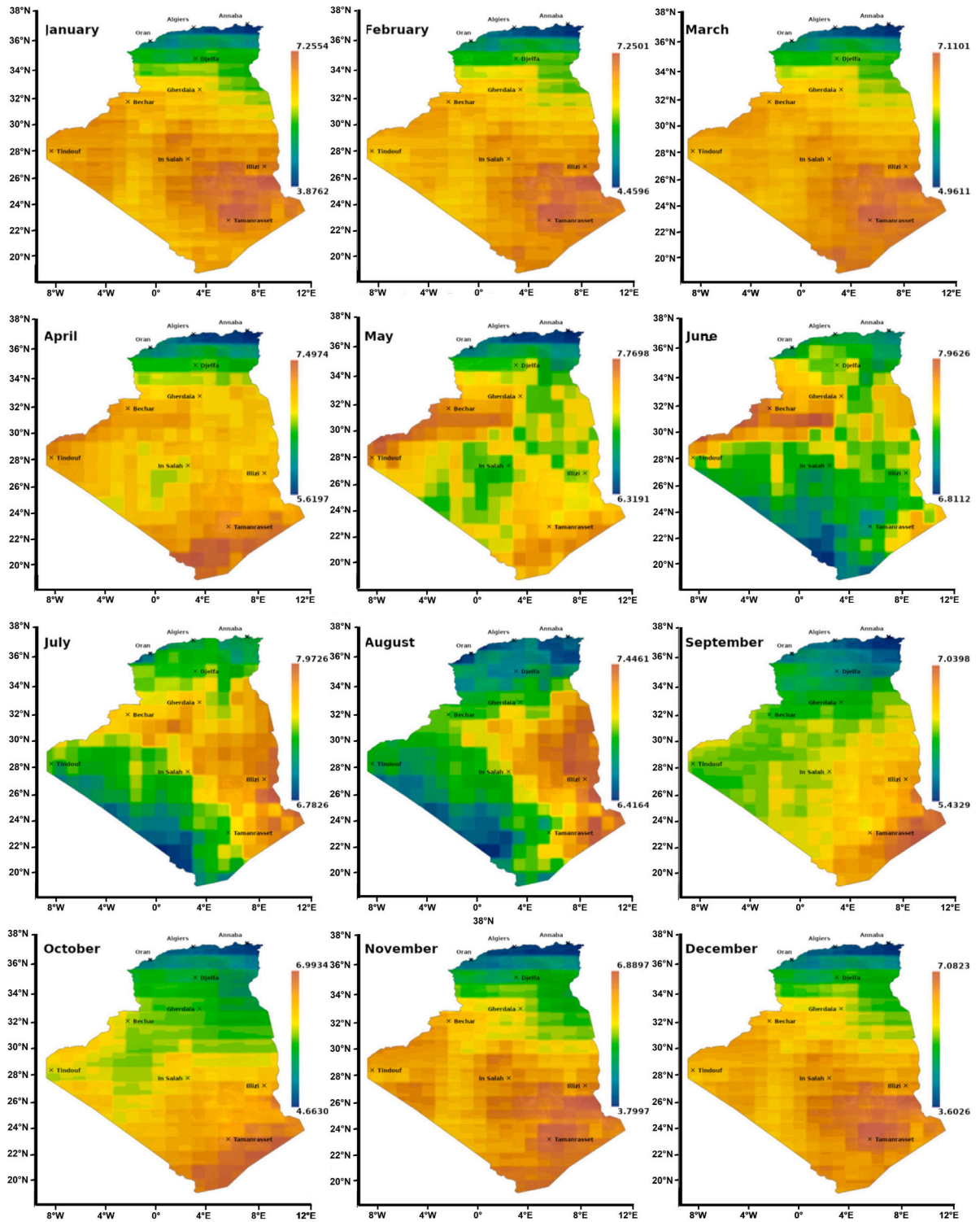


Fig. 11. Monthly mean daily global solar irradiation on optimally tilted surfaces across Algeria. Legend values unit are in kWh/m² day.

- Just as with the parameter $\bar{H}$, it can be observed that the values of the monthly mean daily global solar irradiation on inclined surfaces are higher in the Southern region of Algeria compared to the Northern region.
- The optimal angle of inclination β_{opt} of a surface, which permits the capture of the maximal quantity of solar irradiation on said surface, differs from one region to another and from one month to the next. In Algeria, the value of β_{opt} fluctuates between 0° and 63° over the course of the entire year.

- The optimal value β_{opt} reaches a minimum (as well as a maximum) of 0° in June and July. This can be explained by the fact that during these two months, the sun reaches its highest position in the sky (summer months).
- The value of β_{opt} increases during the winter months and reaches a maximum in December across all regions of Algeria.

The performed linear regression to estimate the relationship between β_{opt} and ϕ gave the results grouped in the Table 9. The value of the correlation coefficient r ranges from -1 to 1. A value equals

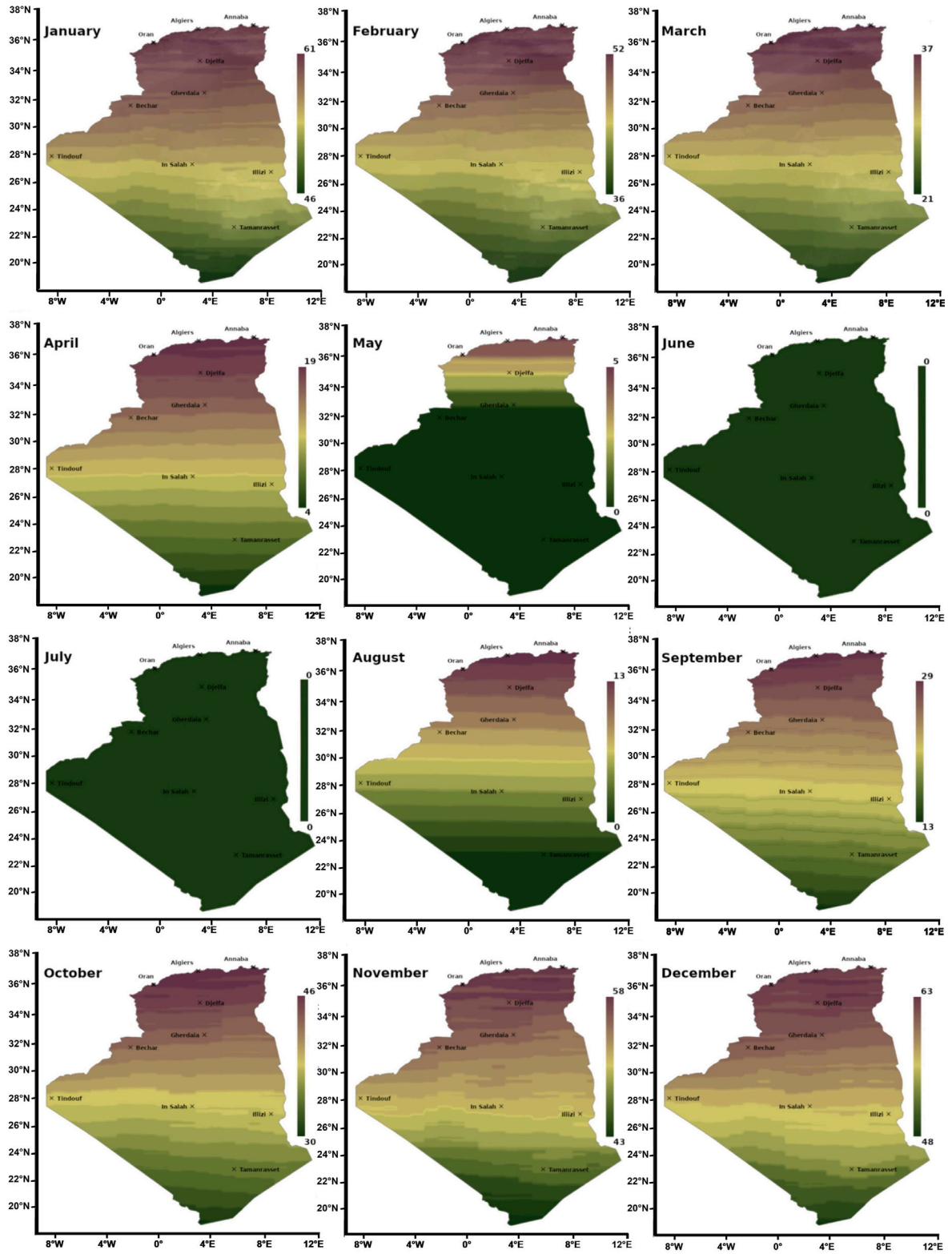


Fig. 12. Optimal tilt values. Legend values unit are in degrees ($^{\circ}$).

to 0 means no relationship (the closer one approaches this value, the weaker the linear relationship), and 1 (or -1) means 100% related. Analyzing the data in Table 9, it is observed that for the majority of the months of the year, except June and July, the value of r approaches 1, indicating a perfect positive correlation: β_{opt} tends to increase as ϕ increases. However, when analyzing the results for the two sunniest months of the year, June and July, the value of r is equal to 0, which

can be explained by the fact that the optimal angle during this period is always equal to 0 (horizontal), and does not depend on the latitude of the location where solar collectors should be installed.

3.3.2. Comparison and validation of results

For validation purposes regarding the obtained results, particularly those concerning the quantity of $\bar{H}$ (as the calculations of $\bar{H}_T$ and β_{opt}

Table 9

Parameters required for calculating the optimal tilt value based on the formula (15).

Month	Parameter a_{β}	Parameter b_{β}	Coefficient of correlation r
January	0.8339322187380801	31.323921093394283	0.9832448144517969
February	0.9086777863420442	19.443200597500425	0.9921490397220463
March	0.8800501852560345	4.8543977860368095	0.9924751276665614
April	0.9052092895723484	-13.386775713985736	0.995243367627949
May	0.18287280764717243	-4.69811657794719	0.6995811212820856
June	0.0	0.0	0.0
July	0.0	0.0	0.0
August	0.8849485333972913	-19.53240055015388	0.9922290431138175
September	0.8919204652809859	-3.496117070752902	0.9954705490705614
October	0.8551544327003304	14.251013288795992	0.9939069444735946
November	0.8513932668860384	27.393048671790996	0.9839145201091788
December	0.798137969733999	34.47745171720955	0.9861241992427684

Table 10Values of $\bar{H}$ for the 48 stations referenced by the Algerian Solar Atlas by CDER (columns "CDER") in comparison with values from HALIMI ET AL. (columns "HAL").

N°	Latitude and Longitude (ϕ, ψ)	Values of $\overline{H}$ (kWh/m ² day)								N°	Latitude and Longitude (ϕ, ψ)	Values of $\overline{H}$ (kWh/m ² day)							
		Winter		Spring		Summer		Autumn				Winter		Spring		Summer		Autumn	
		CDER	HAL	CDER	HAL	CDER	HAL	CDER	HAL			CDER	HAL	CDER	HAL	CDER	HAL	CDER	HAL
01	(27.80, -0.18)	5.1700	4.9724	7.2720	6.9748	6.4800	6.7612	4.4940	4.2920	25	(28.05, 9.63)	5.1700	4.9724	7.2720	6.9748	6.4800	6.7612	4.4940	4.2920
02	(32.76, -0.60)	4.5980	4.5353	7.5160	7.1570	6.8970	6.6368	4.0080	3.8319	26	(27.23, 2.50)	4.5980	4.5353	7.5160	7.1570	6.8970	6.6368	4.0080	3.8319
03	(36.68, 3.22)	3.6050	3.4138	6.6640	6.3800	6.4190	6.3137	2.9170	2.8971	27	(35.17, 2.32)	3.6050	3.4138	6.6640	6.3800	6.4190	6.3137	2.9170	2.8971
04	(36.83, 7.80)	3.4870	3.3020	6.7580	6.2873	6.5670	6.2469	2.8980	2.8134	28	(35.60, 0.30)	3.4870	3.3020	6.7580	6.2873	6.5670	6.2469	2.8980	2.8134
05	(35.82, -0.27)	3.6790	3.7196	6.4980	6.5513	6.1940	6.2763	2.9980	3.1045	29	(34.52, -0.28)	3.6790	3.7196	6.4980	6.5513	6.1940	6.2763	2.9980	3.1045
06	(35.33, 5.33)	4.0270	3.7844	7.3760	6.6771	6.9080	6.3705	3.3200	3.1754	30	(36.28, 2.73)	4.0270	3.7844	7.3760	6.6771	6.9080	6.3705	3.3200	3.1754
07	(35.75, 6.32)	3.8630	3.7244	7.1700	6.5595	6.9600	6.2534	3.3110	3.1548	31	(34.82, -1.78)	3.8630	3.7244	7.1700	6.5595	6.9600	6.2534	3.3110	3.1548
08	(36.07, 4.77)	4.0020	3.3653	7.3320	6.3555	6.9970	6.3375	3.3550	2.8589	32	(36.30, 2.23)	4.0020	3.3653	7.3320	6.3555	6.9970	6.3375	3.3550	2.8589
09	(31.50, -2.25)	4.8110	4.7343	7.8050	7.3476	9.9320	6.7638	4.2050	4.0362	33	(35.88, 0.12)	4.8110	4.7343	7.8050	7.3476	9.9320	6.7638	4.2050	4.0362
10	(36.72, 5.07)	3.4470	3.3531	6.5760	6.3464	6.3290	6.2172	2.8490	2.8407	34	(35.67, 4.50)	3.4470	3.3531	6.5760	6.3464	6.3290	6.2172	2.8490	2.8407
11	(30.13, -2.17)	4.9390	4.8502	7.6440	7.3548	6.8250	6.7729	4.3160	4.1198	35	(33.27, -0.30)	4.9390	4.8502	7.6440	7.3548	6.8250	6.7729	4.3160	4.1198
12	(35.30, -1.35)	3.7940	3.7558	6.7870	6.5894	6.3050	6.2278	3.1370	3.1648	36	(35.63, -0.60)	3.7940	3.7558	6.7870	6.5894	6.3050	6.2278	3.1370	3.1648
13	(34.80, 5.73)	4.3060	4.0748	7.4270	6.7587	6.8380	6.4575	3.5600	3.3811	37	(34.87, 0.15)	4.3060	4.0748	7.4270	6.7587	6.8380	6.4575	3.5600	3.3811
14	(35.33, 4.20)	3.9580	3.8270	7.0370	6.7163	6.6230	6.4018	3.3680	3.1816	38	(36.18, 5.25)	3.9580	3.8270	7.0370	6.7163	6.6230	6.4018	3.3680	3.1816
15	(36.20, 1.33)	3.4630	3.5643	6.6170	6.5229	6.6490	6.2979	2.9870	2.9388	39	(35.20, -0.62)	3.4630	3.5643	6.6170	6.5229	6.6490	6.2979	2.9870	2.9388
16	(36.28, 6.62)	3.6630	3.3407	7.0030	6.3430	6.7900	6.2182	3.0800	2.8353	40	(36.88, 6.90)	3.6630	3.3407	7.0030	6.3430	6.7900	6.2182	3.0800	2.8353
17	(36.92, 3.95)	3.480	3.4138	6.6580	6.3800	6.3250	6.3137	2.8600	2.8971	41	(36.28, 7.97)	3.480	3.4138	6.6580	6.3800	6.3250	6.3137	2.8600	2.8971
18	(24.27, 9.47)	6.1740	5.3481	7.7820	7.0905	7.9790	7.0671	4.9200	4.7761	42	(22.80, 5.43)	6.1740	5.3481	7.7820	7.0905	7.9790	7.0671	4.9200	4.7761
19	(34.33, 3.38)	4.1430	4.0622	7.4310	6.8832	7.0270	6.4389	3.6070	3.3901	43	(35.42, 8.12)	4.1430	4.0622	7.4310	6.8832	7.0270	6.4389	3.6070	3.3901
20	(33.67, 1.00)	4.1650	4.2787	7.4910	7.0092	6.9820	6.5636	3.5910	3.5788	44	(35.35, 1.47)	4.1650	4.2787	7.4910	7.0092	6.9820	6.5636	3.5910	3.5788
21	(30.57, 2.87)	4.9390	4.7290	7.4610	7.3711	6.8720	6.9321	4.2860	3.9668	45	(29.25, 0.28)	4.9390	4.7290	7.4610	7.3711	6.8720	6.9321	4.2860	3.9668
22	(33.50, 6.78)	4.5070	4.1549	7.3020	6.9543	6.8580	6.6501	3.7040	3.4388	46	(27.70, -8.17)	4.5070	4.1549	7.3020	6.9543	6.8580	6.6501	3.7040	3.4388
23	(32.40, 3.80)	4.6960	4.5066	7.5010	7.1188	6.9730	6.7018	3.9780	3.7784	47	(35.02, -1.47)	4.6960	4.5066	7.5010	7.1188	6.9730	6.7018	3.9780	3.7784
24	(36.47, 7.47)	3.5600	3.3020	6.7290	6.2873	6.6800	6.2469	2.9750	2.8134	48	(33.07, 6.08)	3.5600	3.3020	6.7290	6.2873	6.6800	6.2469	2.9750	2.8134

are essentially based on this quantity), the Algerian Solar Atlas of the CDER [27] was utilized. This entity possesses access to comprehensive datasets from the ONM, ensuring that its results reflect the most reliable solar radiation data available in Algeria. By comparing the findings with those of the CDER, the credibility of the research is reinforced, aligning with the authoritative standards in the field.

After analyzing Fig. 13, which reproduces the CDER maps concerning the monthly average of the global irradiation received on a horizontal surface (a map for each season), the calculations have been redone, resulting in the generation of new seasonal maps. It has been observed that there is a high degree of similarity between the solar potential data provided by the CDER and those generated in this study. However, a comparative analysis between the maps illustrated in Fig. 13, indicates that the methodology presented has the advantage of producing more precise results, particularly in providing a more detailed view of the distribution of the solar resource across the Algerian territory (this is largely due to the fact that the calculations have been made using a larger sample of data, covering the period from 01/01/1984 until 31/12/2021, totaling 38 years).

In the case of the mean absolute percentage error, a lower percentage corresponds to a more favorable outcome. Therefore, the *MAPE* values for the 48 stations cited in the Solar Atlas (Table 10) have been calculated. The results for each station are illustrated in Fig. 14. An analysis of these results reveals that:

- For the majority of the selected stations and during the 4 seasons of the year, the *MAPE* generally does not exceed the value of 16%, except for station n°38 where it reaches 18.02% during the winter, and station n°9 where the *MAPE* is equal to 31.9% during the summer.
- Among the 48 selected stations, 44 have a *MAPE* of less than 10% during the winter season, 46 during the spring season, 42 during the summer season, and 45 during the autumn season.

Regarding the overall *MAPE* values (a single value per season for all stations), they do not exceed 6% throughout the entire year (4.96% for winter, 5.68% for spring, 5.34% for summer, and 4.66% for autumn). All these results confirm, once again, the accuracy of the methodology and thus facilitate its validation.

3.4. Daylight and sunshine duration maps

Fig. 15(a) and (b) illustrate the annual daily average sunshine duration using the SVR and LRM models, respectively. These maps show the distribution of sunshine estimation for all coordinates in Algeria. Therefore, serving as a comprehensive representation of the solar exposure across this country. When examining the two figures collectively and assessing their *MAPE* values (Fig. 16) in relation to the experimental data as provided by the Algerian Solar Atlas [27], it becomes apparent that the support vector regression model has exhibited superior predictive performance compared to the linear regression models. Globally for all the stations, the *MAPE* for the SVR stands at 15.17%, whereas the *MAPE* for the LRM is notably higher at 43.92%.

Among the results of Fig. 15, the utilization of the SVR method reveals an estimated total annual value of approximately 3597 hours for sunshine duration (average between the maximum and minimum throughout the entire year). Notably, this value falls within a similar range as the determination made by the Ministry of Energy and Mines (3500 hours of sunshine annually [6]).

Fig. 17 depicts the total annual duration of daylight measured in hours. Its maximum duration reaches approximately 4553 hours per year, surpassing the total annual sunshine duration. This disparity arises from the fact that daylight hours encompass the entirety of natural light throughout the day, encompassing both direct sunlight and indirect light during twilight periods. In contrast, sunshine hours specifically denote the period when direct sunlight is present.

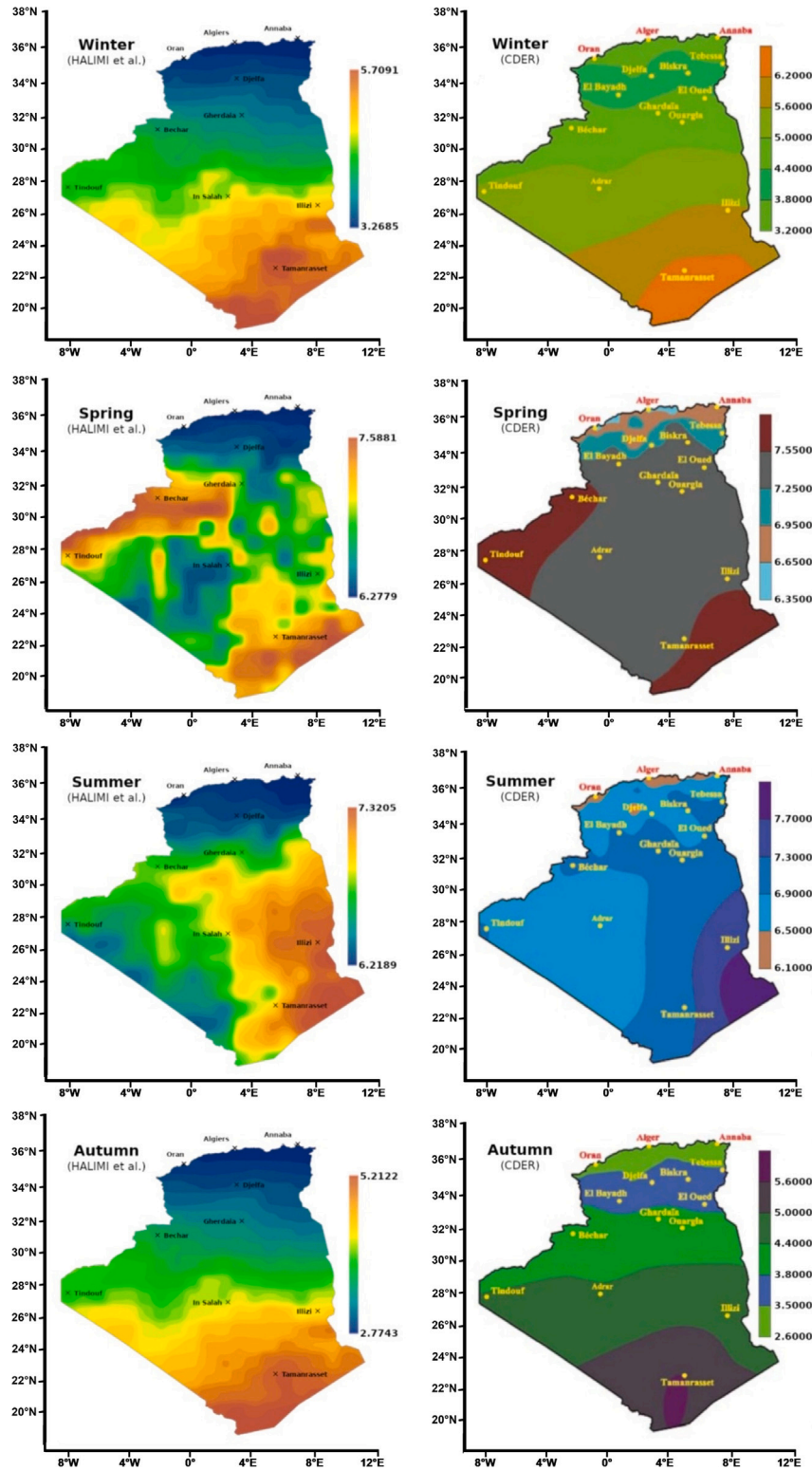


Fig. 13. Seasonal comparison of global horizontal irradiation data from Halimi et al. maps and the CDER maps. Legend values unit are kWh/m^2 day.

The findings demonstrate a close agreement between the estimated values and the CDER experimental data. The SVM model, in particular, exhibited smaller errors and better accuracy compared to the LRM model. This validates the effectiveness of the approach in estimating sunshine duration based on the given parameters.

3.5. Analyzing Algeria's solar PV potential and site suitability

3.5.1. Solar PV potential in Algeria

Using Eq. (16), Python programming language and GIS tools, a map that summarizes the potential for photovoltaic solar energy production in Algeria has been drawn (Fig. 18(a)). This map represents the annual sum of the photovoltaic energy E_{pv} produced by a PV power plant with

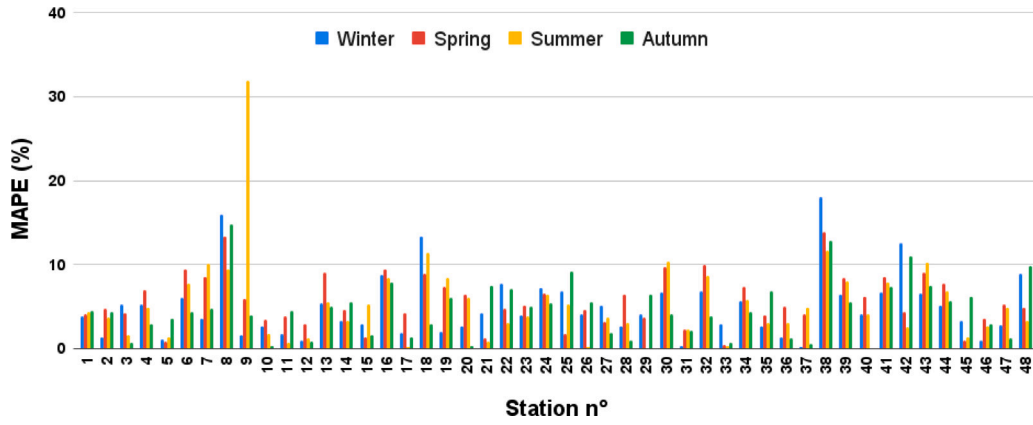


Fig. 14. Mean absolute percentage error for the global horizontal irradiation results compared to those from the Solar Atlas by CDER.

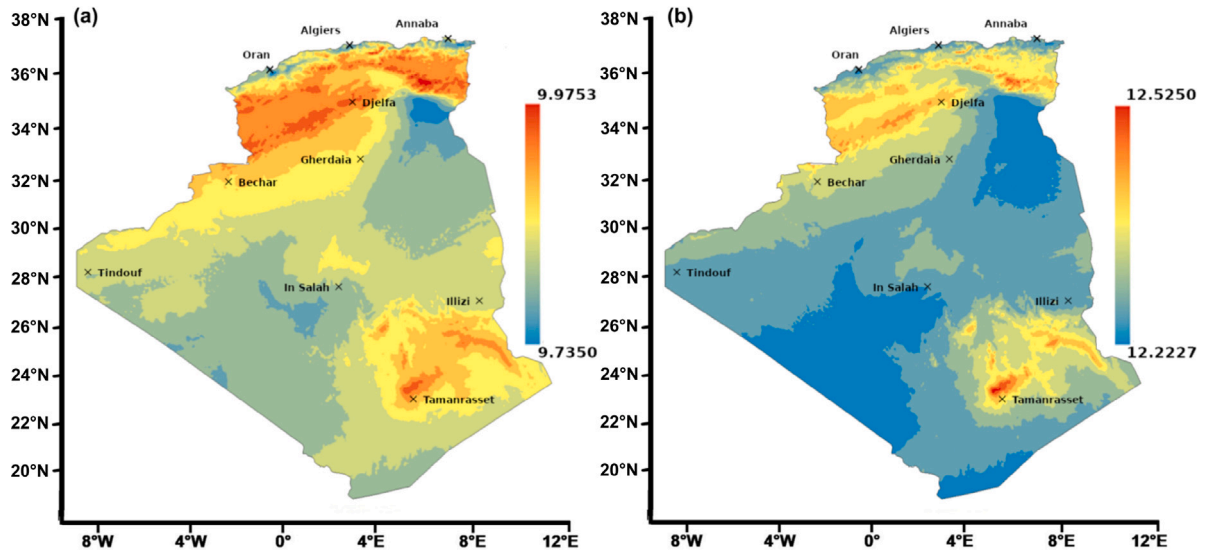


Fig. 15. Annual daily average sunshine duration using the (a) SVR model and (b) the LRM model. Legend values represent time in hours.

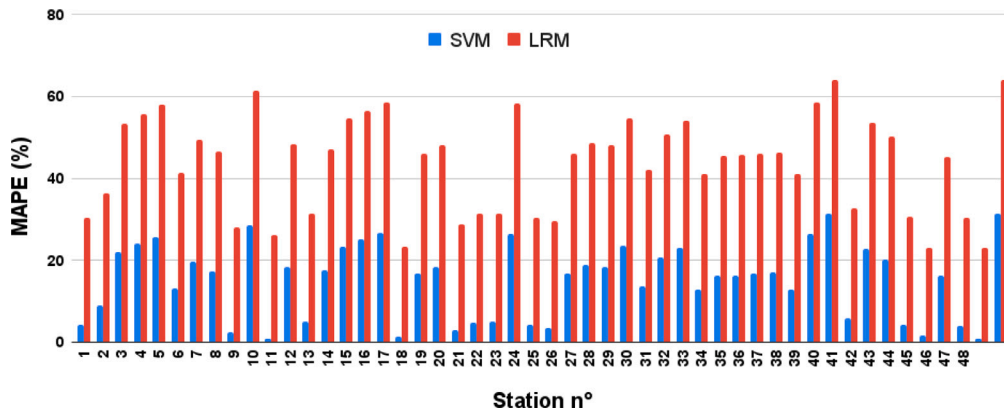


Fig. 16. Mean Absolute Percentage Error for sunshine duration results compared to those from the Solar Atlas by CDER.

a total installed capacity of 1 MWp and whose panels are oriented due south (fixed tilt angle β equal to ϕ), throughout the year. The necessary details for the calculations concerning the type of photovoltaic modules used are grouped in Table 11.

The presented Fig. 18 shows two maps of the produced photovoltaic potential in Algeria. The map on the left depicts the total annual E_{pv} in MWh/MWp, as calculated in this study, and the map on the right

represents the CDER assessment of the same parameter (by [28]). Upon conducting an analysis, it is noted that:

- The map presented shows a range of E_{pv} values from 1535 MWh/MWp to 2051 MWh/MWp per year, while the CDER map indicates a wider range from 1650 MWh/MWp to 2450 MWh/MWp per year. This difference can be attributed to the selection of solar panel parameters, as the analysis assumed a

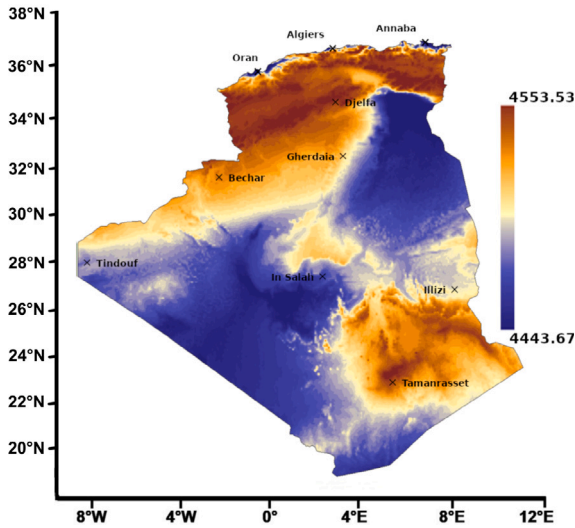


Fig. 17. Annual sum daylight hours. Legend values represent time in hours.

Table 11
Characteristics of photovoltaic panels used.

Monocrystalline 72-Cell photovoltaic panel from the Condor electronics brand [123]			
Solar cell surface	P_{max}	$\eta_{pv,ref}$	$\alpha_{pv,max}$
125 × 125 mm ²	200 W _p	17.80%	−0.41 % °C ^{−1}

photovoltaic efficiency of 17.8% based on a model of a commercially available panel in Algeria. In contrast, the CDER assessment utilized a panel with an efficiency of 19%, without specifying the source of the PV module model employed. Furthermore, based on the formula (17), the calculated power loss coefficient C_p varies throughout the year ($C_p = 1 - K$), in contrast to the fixed value of 26% utilized in the CDER study.

- The map in Fig. 18(a) reveals that the highest E_{pv} values are located in Southeast Quadrant, particularly in East–Southeast (ESE) and to a lesser extent in South–Southeast (SSE) regions. Previous results have identified Southeast Quadrant as the hottest region in Algeria, with high global horizontal irradiance and global tilted irradiance values throughout the year. Medium E_{pv} values are generally distributed across both the Southeast and Southwest Quadrants, while lower photovoltaic production is observed in the Northeast and Northwest Quadrants. The lowest values, depicted in a distinct blue color on the map, are predominantly concentrated in the Northeast Quadrant and some centered coastal regions.
- The CDER map indicates relatively high E_{pv} values in the smaller areas near Djanet and Tamanrasset, which differs from the findings of a more widespread distribution of the highest E_{pv} values along the edges of the study region. Specifically, the Western edge above Tindouf is entirely within the red color range, indicating E_{pv} values of approximately 2350 MWh/MWp to 2450 MWh/MWp. In contrast, the map on the left shows a more gradual transition in this region, with medium values ranging from yellow to darker yellow, interspersed with lower values. This uneven distribution is not evident in the CDER map. Furthermore, the lowest E_{pv} values in the CDER map are distributed along the Algerian coastline, whereas the alternative map indicates these lower values concentrated in a smaller area in the North–Northeast (NNE) region.

Table 12
Climate type distribution in regions by PV suitability grades.

Grade	BWh	BWk	BSh	BSk	Csa
Minimally	0	0	0	8.84%	91.36%
Marginally	15.74%	1.67%	2.86%	12.37%	67.50%
Somewhat	60.92%	2.97%	6.05%	3.02%	28.83%
Moderately	86.15%	2.77%	3.12%	0.39%	8.74%
Very	96.67%	1.60%	1.31%	0.01%	3.16%
Highly	99.17%	0.68%	0.41%	0	0.41%
Extremely	98.10%	1.16%	0	0	0.36%

Overall, the CDER map appears to reflect a more generalized, higher-level assessment, while the produced map in this research provides a more detailed, location-specific evaluation of photovoltaic potential.

3.5.2. Evaluation of PV implementation sites using multiple criteria analysis

Fig. 19 depicts the resultant map after the reclassified and weighted process, which was undertaken to obtain the suitability map shown in Fig. 20. This was the product of the complete integrated approach described generally in Fig. 7. Analyzing Fig. 20 indicates that areas within grade 9 are classified as extremely suitable (prime sites) for PV system installation, offering the highest potential for energy generation and return on investment. Conversely, grade 1 refers to unsuitable sites. The areas falling within the extremely suitable to very suitable grades (9 to 7) have been prioritized when applying the AHP method, in contrast to the lower grades where areas have been given lower priority to avoid. On top of that, the AHP methodology allowed to obtain a structured and analytical approach to evaluating multiple criteria and arriving at a prioritized set of factors that can guide site selection.

Comparing Fig. 20 to Fig. 18(a), with the parameter E_{pv} which has the highest percentage factor of 36%, indicates that the results are in good accordance. Extremely and highly suitable sites are predominantly located in a non-homogeneous distribution somewhat between the ordinal directions east (E) and south (S), with additional highly suitable sites found along the edges of the Northwest Quadrant (in West–Northwest). Very suitable areas are generally found with a higher concentration in both Northwest and Southwest Quadrants, while moderately suitable areas are found in the Southwest Quadrant, interspersed with a few areas categorized as somewhat suitable.

The Northeast Quadrant of Algeria is predominately occupied by somewhat suitable areas. This last grade is also found in the highland and littoral surface, intercepted by two other classes being marginally and minimally suitable areas. When analyzing the entire length of the Algerian coastline, it becomes evident that these areas are generally concentrated in the Northeast Quadrant of the littoral.

The following data presents a land suitability analysis, detailing both the specific area and the percentage of Algeria's total area for each suitability category:

Extremely Suitable : 5476 km² (0.2463%).

Highly Suitable : 46749 km² (2.0042%).

Very Suitable : 263296 km² (11.2877%).

Moderately Suitable : 757194 km² (32.4614%).

Somewhat Suitable : 849137 km² (36.4030%).

Marginally Suitable : 370025 km² (15.8632%).

Minimally Suitable : 40452 km² (1.7342%).

The results illustrated in Fig. 20 show the exact coordinates of the optimal sites for PV system installation across Algeria (for the regions, see Fig. 21). Moreover, the site selection analysis clearly demonstrates that it is closely linked to the spatial distribution of solar irradiance across the country as illustrated in Fig. 10. Concerning the climate maps (Fig. 9), the H1 climate zone is generally characterized by a mix of somewhat, marginally, and minimally suitable sites, with a very reduced number of points which represents moderately suitable grade. The latter occupies also the H2 climate zone with somewhat suitable

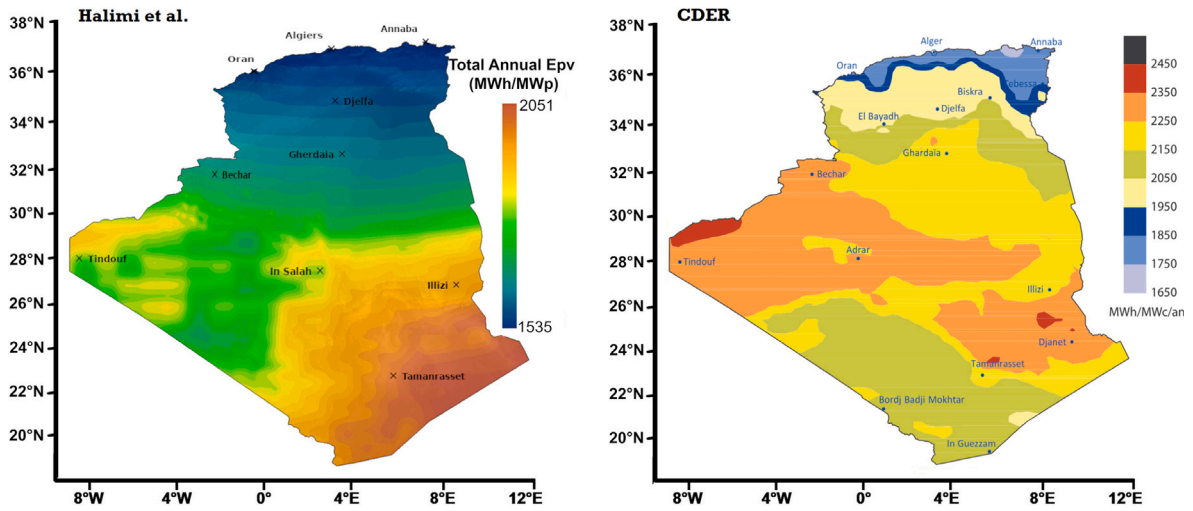


Fig. 18. Annual total photovoltaic energy (E_{pv}): (a) Data from HALIMI et AL., (b) Data from CDER.

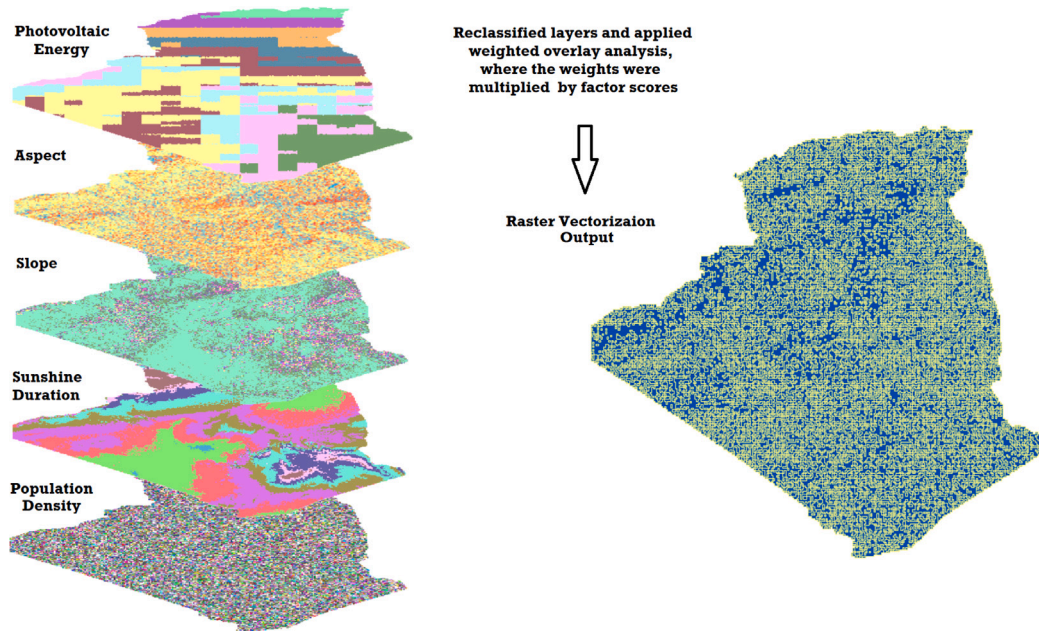


Fig. 19. Rasterization of the reclassified and weighted layers.

areas (more prevalent in the east). For the Sahara H3 climate zone, areas between 669.58 – 2845 m are extremely suitable, while –33 to 2442 m have a mix of highly, very, moderately, and somewhat suitable sites for PV. Finally, referring to the Köppen–Geiger classification, it is evident that the extremely suitable sites are located particularly in a desert, arid and hot climatic zone (BWh) (Table 12).

3.5.3. Comparison of results and improvements made by the study

Studies aiming at producing a high-resolution suitability map showing the spatio-temporal variability of PV potential across the entire Algerian territory are very rare. In reviewing the literature, as previously presented in the introduction part of this article, SETTOU's research work [21] has been considered to date as the most comprehensive among other studies. Authors of [21] employed a multi-criteria decision-making approach incorporating seven parameters for the site selection of PV systems in Algeria. These parameters included proximity to the electric grid, maximum solar resource (GHI), proximity to major roads, proximity to urban areas, minimization of soil water content, slope, and maximum distance to land use. While this approach

provides a broad overview, the off-grid solar assessment study adopts a more focused methodology by selecting five key criteria that significantly influence solar energy potential: annual sum of photovoltaic energy, sunshine duration, slope, aspect, and population density. The rationale for this selection is grounded in the direct relationship of climate factors, such as annual solar radiation and sunshine duration, with energy production. Topographical elements, notably aspect and slope, determine the efficiency of solar panel installation, while demographic factors, particularly population density, help identify areas with high energy demand. Notably, the exclusion of proximity to roads was implemented to maximize solar photovoltaic potential across all regions, including remote areas, thereby facilitating both large-scale and residential solar projects. This comprehensive approach not only promotes sustainability and reduces reliance on fossil fuels but also fosters local economic empowerment through job creation and reduced energy costs.

Furthermore, in the context of solar irradiation mapping, most studies conducted do not employ a sufficiently extensive array of coordinates to encompass the entirety of Algerian territory. Instead,

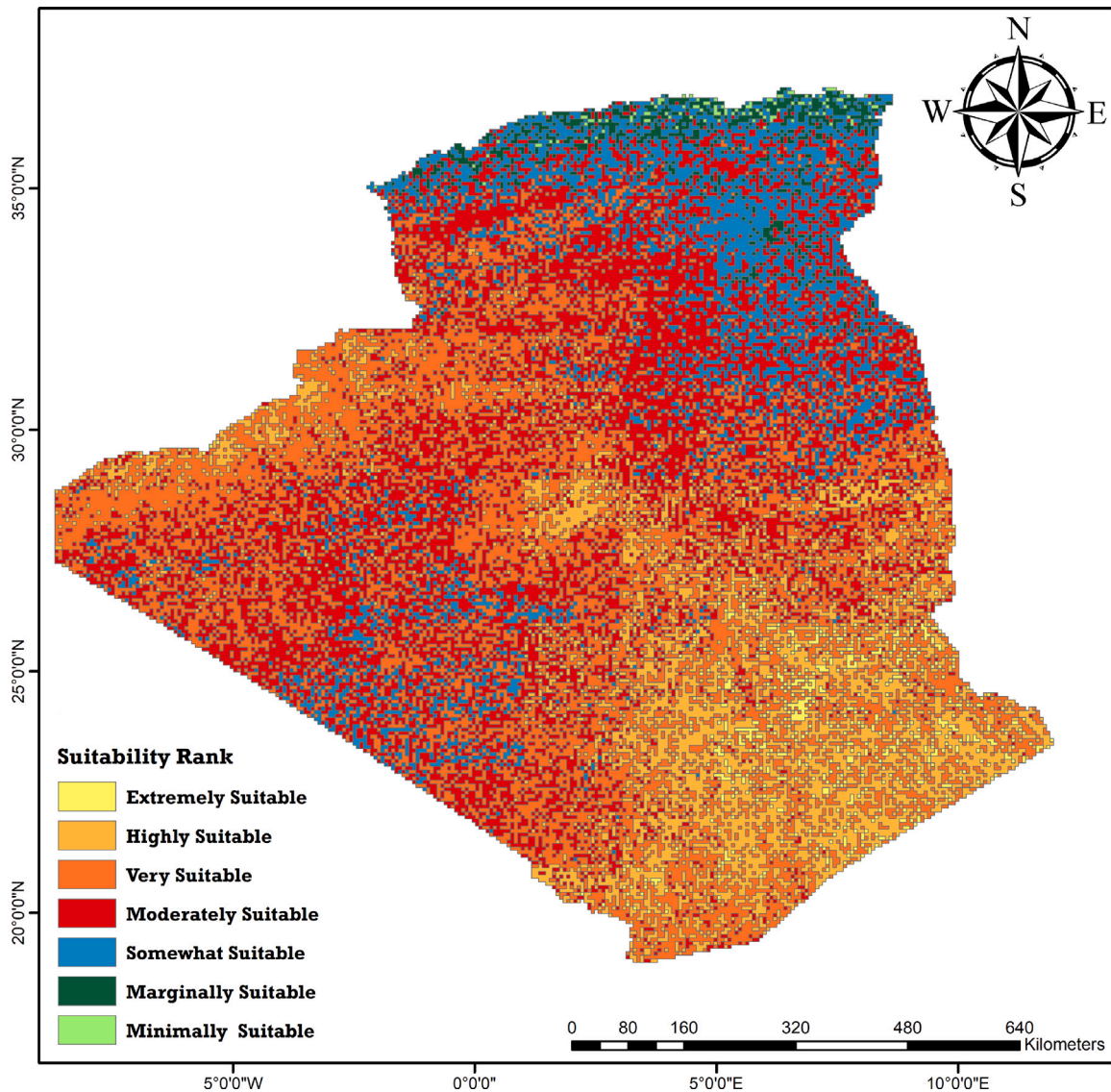


Fig. 20. Suitability map for photovoltaic implementation in Algeria.

they tend to rely on a limited number of meteorological stations, which are frequently located considerable distances apart, particularly in the Southern regions of the country. For example, SETTOU et AL. [21] initially utilized a solar irradiation map derived from 64 data points, later expanding to 216 points to plot their final solar irradiation map. This limited data scope may lead to discrepancies in solar resource estimations across different regions. In contrast, the produced solar irradiation map in this study, based on a comprehensive dataset covering the entirety of Algeria, reveals significant differences in solar potential.

Regarding slope analysis, the resolution of the slope map adopted by SETTOU et AL. [21] is notably lower than the one used in this study. This discrepancy can be attributed to the less accurate digital elevation model employed in their research (the vast majority of studies rely on NASA's DEM, or another less precise one), which may compromise the overall reliability of the site selection process. They defined suitable sites as having a slope of less than 10° , while this study identified a slope range of 0° to 0.663956° as most influential, as at near-zero slopes solar PV panel mounts can be adjusted to achieve optimal angles for different seasons and latitudes. Additionally, an aspect map was incorporated into the analysis, which is crucial for understanding sunlight exposure and wind patterns.

In evaluating the resultant suitability map from SETTOU et AL. [21] (Fig. 22), it is evident that the analysis was limited to a selected number of areas, yielding only a few suitability results (especially for grid-connected solar PV systems). Another limitation lies in their reliance solely on *GHI* without incorporating photovoltaic energy production data, which could lead to more realistic site suitability assessments as it is shown in this study. Their classification system categorized areas into five suitability levels: least, marginally, moderately, highly, and most, corresponding to 4.29%, 29.86%, 43.44%, 20.56%, and 1.85% of the study area, respectively. In contrast, this research's findings indicate that minimally suitable areas account for only 1.73% ($40,452 \text{ km}^2$) of the total surface area, while the most suitable regions representing 0.24% ($5,476 \text{ km}^2$), with energy production estimates ranging from 1,984,252,800 to 2,051,221,888 Wh/m^2 per year. This comparative analysis underscores the robustness of the used methodology and highlights the improvements and advantages of a more precise and targeted approach in assessing solar PV implementation sites in Algeria.

4. Conclusion

This study delved into the intricate process of identifying suitable locations for photovoltaic (PV) solar power plants, a task that involves several criteria.

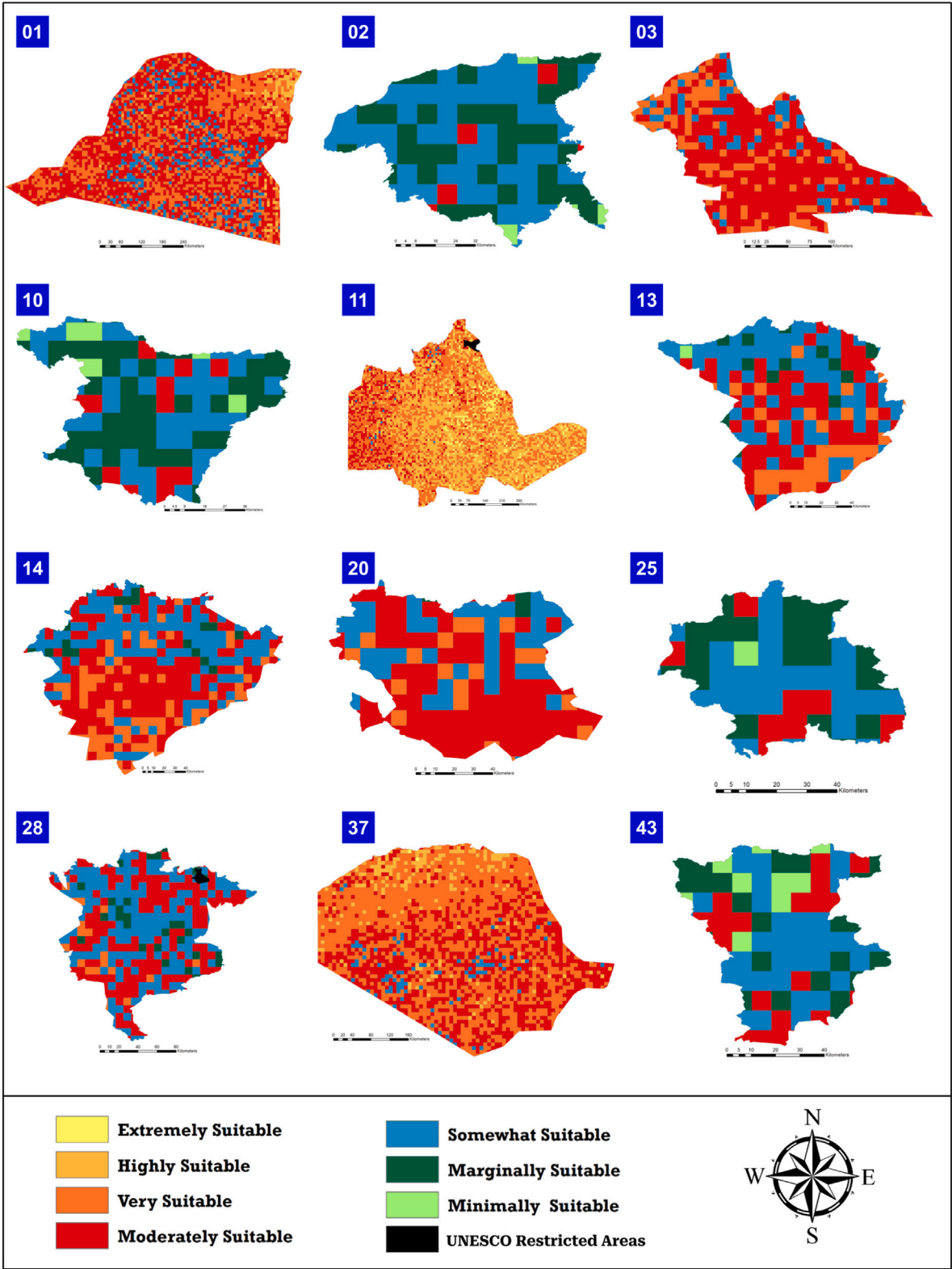


Fig. 21. Suitability maps for photovoltaic implementation in the provinces of Algeria. Please refer to Fig. 1(c) for the exact geographic location of each province.

The findings revealed that Algeria, with its promising potential in solar energy production, has intensified efforts to leverage this advantage. To assist the government and stakeholders in installing PV systems in Algeria, this research introduced a methodology blending

Geographic Information System (GIS) analysis, multi-criteria decision-making (MCDM) techniques, and off-site measurements to pinpoint optimal sites for off-grid PV facilities. Prior to achieving this goal, a refined digital elevation model (DEM) was meticulously developed.

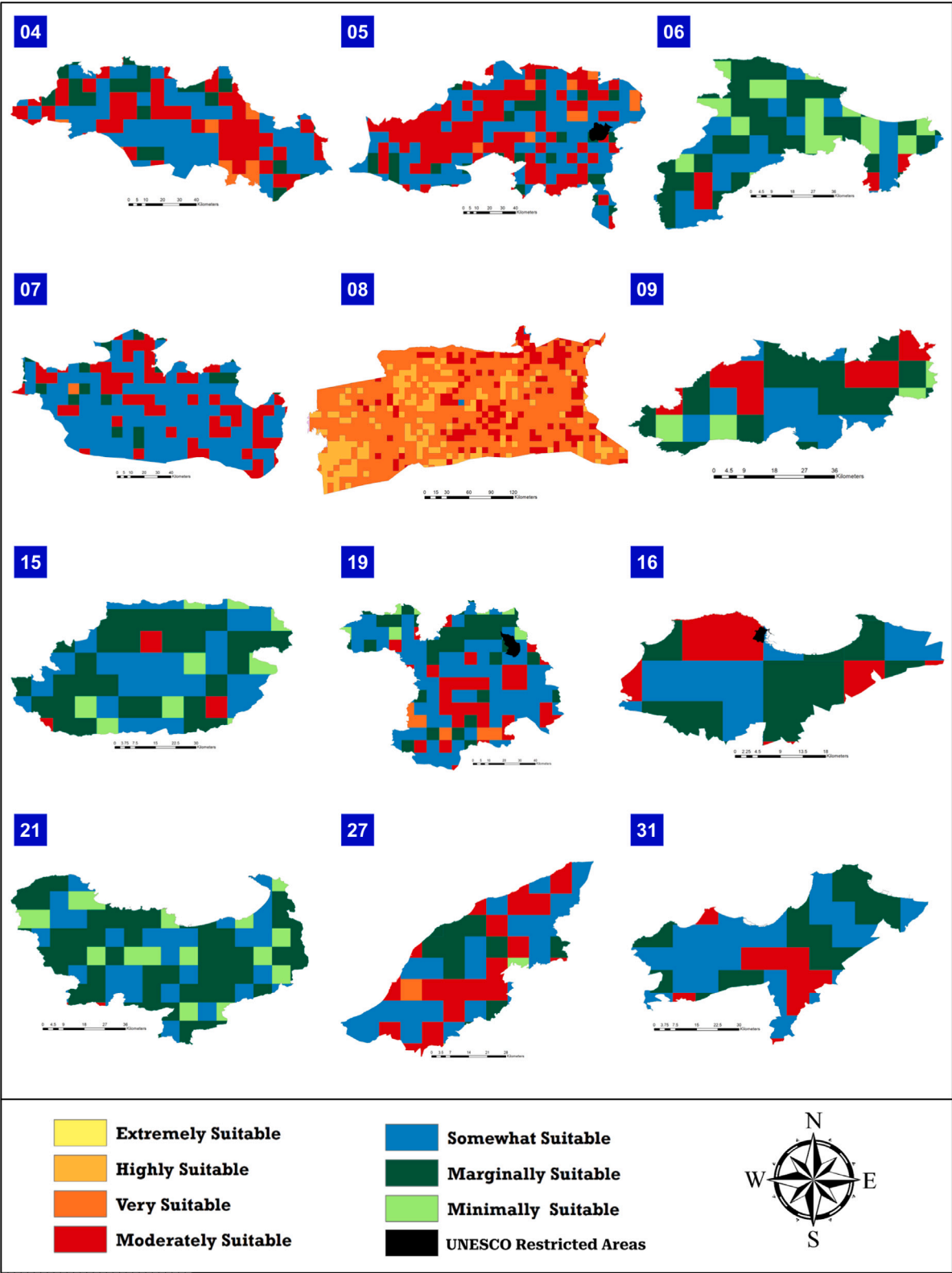


Fig. 21. (continued).

Moreover, monthly average global daily horizontal and tilted irradiance values were calculated for a Typical Meteorological Year (TMY), aiding in mapping these values across the country, with higher concentrations

in Southern regions. The study also estimated the annual photovoltaic energy output from a 1 MWp PV power plant. Utilizing the Analytic Hierarchy Process (AHP), various criteria were assessed to prioritize

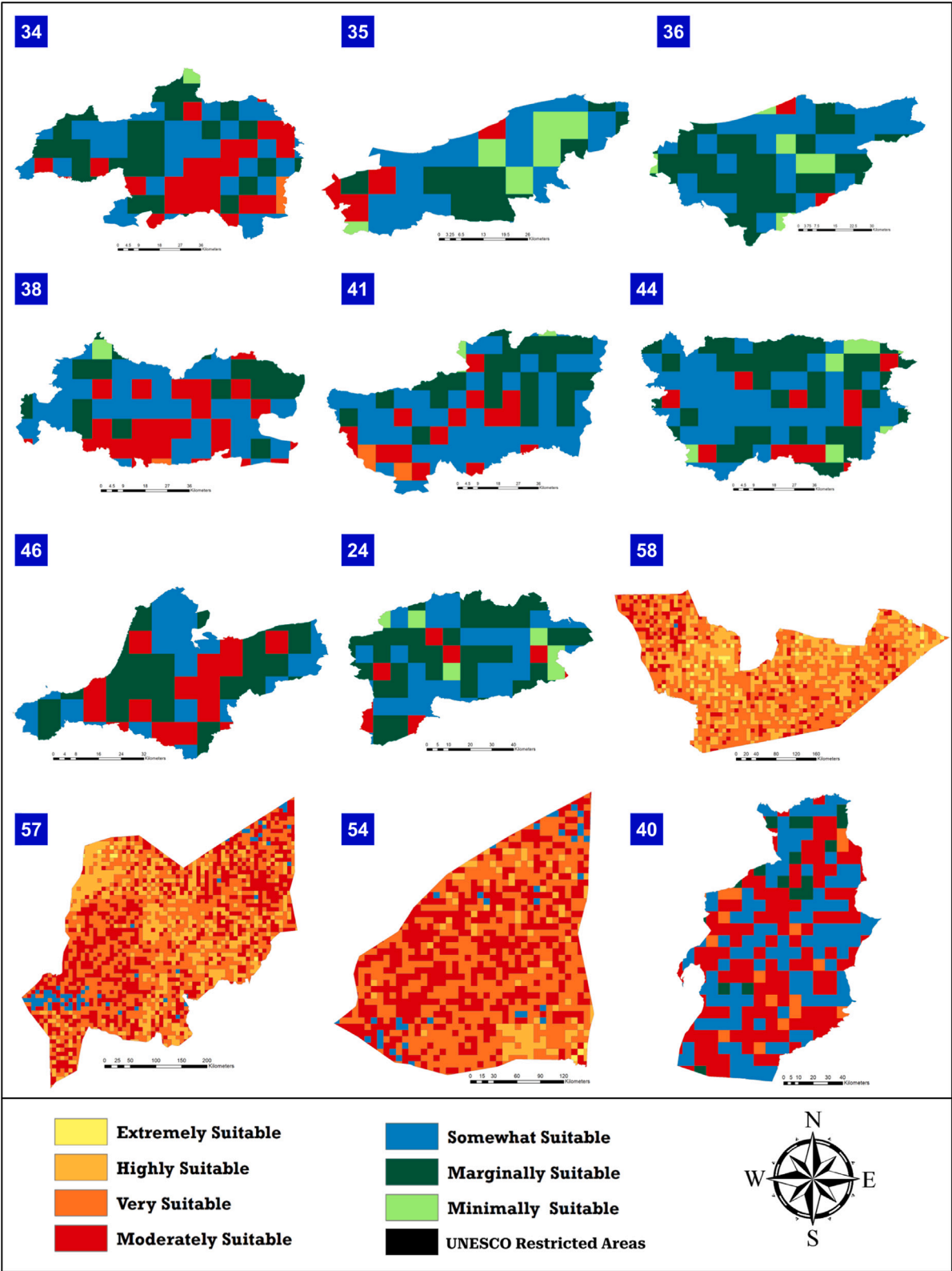


Fig. 21. (continued).

areas, resulting in Algeria being classified into seven grades, with “Somewhat suitable” covering the largest portion at 36.4030%. Therefore, this research produced an updated Solar Atlas of Algeria and a detailed suitability map showcasing the spatial and temporal variability of PV potential nationwide.

The study supports Algeria’s initiative for energy diversification and aims to reduce dependence on fossil fuels, with a target of achieving a 27% share of renewable energy by 2030. By focusing on remote areas, we emphasize decentralized solar solutions that not only benefit local communities but also lower energy costs. The data-driven

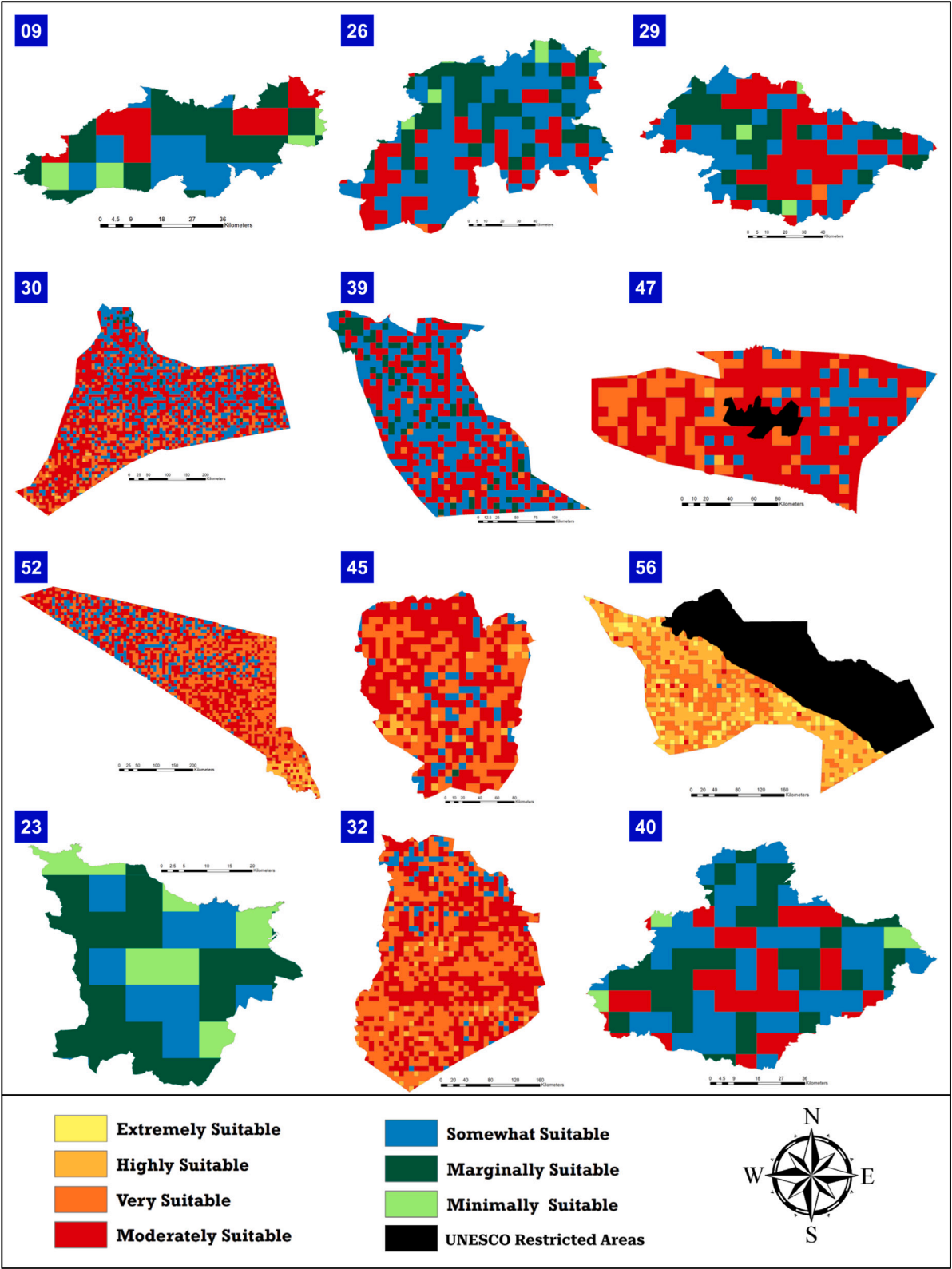


Fig. 21. (continued).

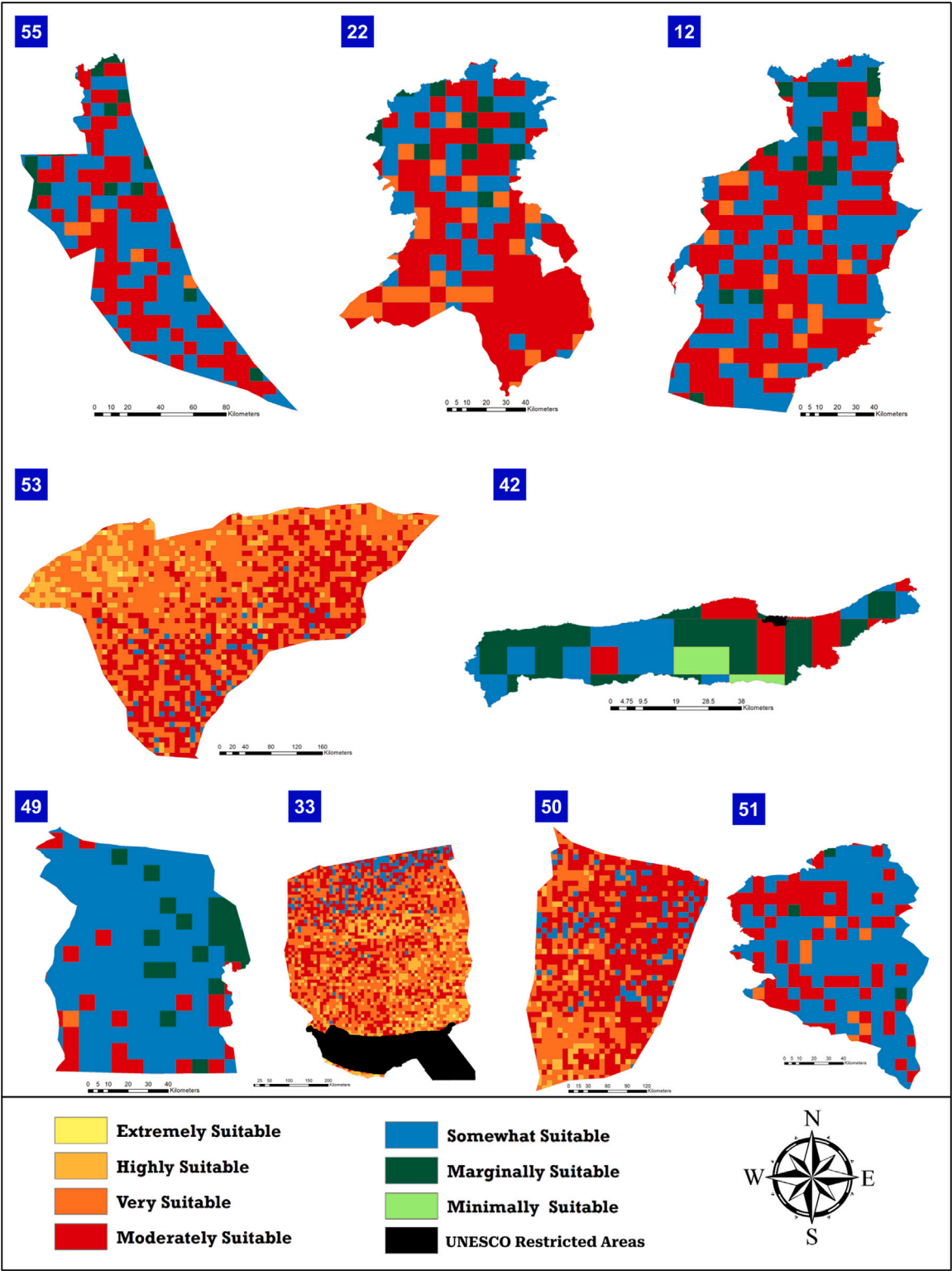


Fig. 21. (continued).

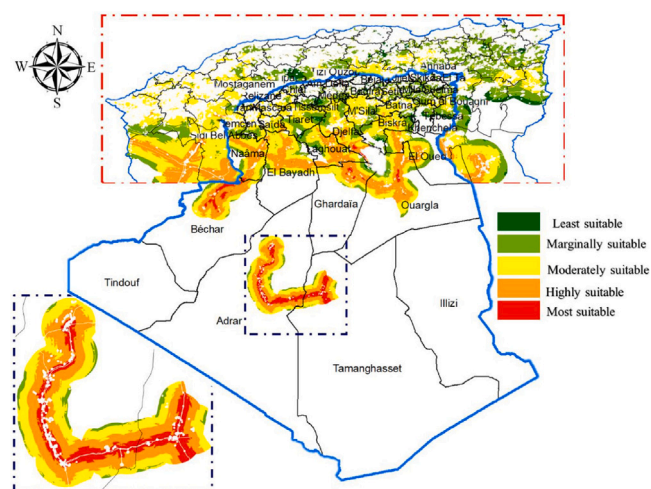


Fig. 22. Suitability Index results using AHP weights from SETTOU ET AL. [21].

analysis informs policymaking and addresses regional energy access disparities.

These findings can assist Algeria in achieving global climate goals while promoting sustainable energy practices. The comprehensive methodology employed in this study could serve as a model for other countries looking to harness solar resources effectively. Ultimately, this research can help Algeria align with international climate commitments and cultivate a sustainable energy landscape. However, the omission of local roads, residential streets, and preservation data due to incomplete references highlights the need for further investigation. Future research should incorporate socioeconomic factors, environmental assessments, and detailed maps of buildings to enhance selection processes and improve connectivity.

CRedit authorship contribution statement

Y. Halimi: Writing – review & editing, Writing – original draft, Visualization, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **S. Halimi:** Software, Resources, Project administration, Investigation. **Z. Bouzid:** Writing – review & editing, Validation, Supervision, Methodology, Conceptualization. **N. Ghellai:** Supervision, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

References

- [1] Nicola M, Alsafi Z, Sohrabi C, Kerwan A, Al-Jabir A, Iosifidis C, Agha M, Agha R. The socio-economic implications of the coronavirus pandemic (COVID-19): A review. *Int J Surg* 2020;78:185–93.
- [2] Werth A, Gravino P, Prevedello G. Impact analysis of COVID-19 responses on energy grid dynamics in Europe. *Appl Energy* 2021;281:116045.
- [3] Balsalobre-Lorente D, Sinha A, Murshed M. Russia-Ukraine conflict sentiments and energy market returns in G7 countries: Discovering the unexplored dynamics. *Energy Econ* 2023;125:106847.
- [4] Foster R, Ghassemi M, Cota A. Solar energy: renewable energy and the environment. CRC Press; 2009.
- [5] eddine Boukela T, Mecibah M-S. Parabolic trough solar thermal power plant: Potential, and projects development in Algeria. *Renew Sustain Energy Rev* 2013;21:288–97.
- [6] Gairaa K, Khellaf A, Benkacilai S, Guermoui M. Solar radiation measurements in Algeria: case of Ghardaia station as member of the enerMENA meteorological network. *Energy Procedia* 2017;141:50–4.
- [7] Group O. The report: Algeria 2011. Oxford Business Group; 2011, URL <https://books.google.dz/books?id=uSGzIPz8cYIC>.
- [8] Fares T, Djeddi A, Nouioua N. Solar energy in Algeria between exploitation policies and export potential. In: 2019 1st international conference on sustainable renewable energy systems and applications. ICSRESA, IEEE; 2019, p. 1–5.
- [9] Bouznit M, Pablo-Romero MdP, Sánchez-Braza A. Measures to promote renewable energy for electricity generation in Algeria. *Sustainability* 2020;12(4):1468.
- [10] Bouraiou A, Necaibia A, Boutasseta N, Mekhilef S, Dabou R, Ziane A, Sahouane N, Attoui I, Mostefaoui M, Touaba O. Status of renewable energy potential and utilization in Algeria. *J Clean Prod* 2020;246:119011.
- [11] Statista. Solar power capacity in Algeria from 2012 to 2022. 2024, <https://www.statista.com/statistics/1370923/solar-energy-capacity-in-algeria/#statisticContainer>, [Online; Accessed 14 December 2023].
- [12] Commission AU, et al. Africa's development dynamics 2022 regional value chains for a sustainable recovery: Regional value chains for a sustainable recovery. OECD Publishing; 2022.
- [13] Moumni A, Hamani N, Moumni N, Mokhtari Z. Estimation du rayonnement solaire par deux approches semi empiriques dans le site de biskra. *Une* 2006;3(4).
- [14] Benkacilai S. Le gisement solaire à Ghardaia. *Bull Énergies Renouvelables* 2007;11.
- [15] Bouzid Z, Debbab M, Belaoui M, Ghellai N. Réactualisation de l'évaluation du gisement solaire et analyse de sa complémentarité avec l'énergie éolienne en algérie. 2016.
- [16] Hassaine L, Mraoui A. Potential of solar electricity for grid-connected systems in Algeria. In: Mediterranean green buildings & renewable energy: selected papers from the world renewable energy network's med green forum. Springer; 2017, p. 561–73.
- [17] Kherbiche Y, Ihaddadene N, Ihaddadene R, Hadji F, Mohamed J, Beghidja A. Solar energy potential evaluation. Case of study: M'sila, an Algerian province. *Int J Sustain Dev Plan* 2021;16(8):1501–8.
- [18] Takilalte A, Harrouni S, Mora J. Forecasting global solar irradiance for various resolutions using time series models-case study: Algeria. *Energy Sour A: Recov Util Environ Effects* 2022;44(1):1–20.
- [19] Benatallah D, Benatallah A, Bouchouicha K, Nasri B. Estimation of clear sky global solar radiation in Algeria. *AIMS Energy* 2019;7(6).
- [20] Bellaoui M, Bouchouicha K, Oulimar I. Estimation of daily global solar radiation based on MODIS satellite measurements: The case study of adrar region (Algeria). *Measurement* 2021;183:109802.
- [21] Settou B, Settou N, Gouareh A, Negrou B, Mokhtara C, Messaoudi D. A high-resolution geographic information system-analytical hierarchy process-based method for solar PV power plant site selection: a case study Algeria. *Clean Technol Environ Policy* 2021;23:219–34.
- [22] Haddad B, Díaz-Cuevas P, Ferreira P, Djebli A, Pérez JP. Mapping concentrated solar power site suitability in Algeria. *Renew Energy* 2021;168:838–53.
- [23] Gouareh A, Settou B, Settou N. A new geographical information system approach based on best worst method and analytic hierarchy process for site suitability and technical potential evaluation for large-scale CSP on-grid plant: An application for Algeria territory. *Energy Convers Manage* 2021;235:113963.
- [24] Díaz-Cuevas P, Haddad B, Fernandez-Nunez M. Energy for the future: Planning and mapping renewable energy. The case of Algeria. *Sustain Energy Technol Assess* 2021;47:101445.
- [25] Capderou M. Atlas solaire de l'Algérie. Office des publications Universitaires; 1986.
- [26] Mefti A, Bouroubi Y, Khellaf A. Analyse critique du modèle de l'atlas solaire de l'Algérie. *J Renew Energies* 1999;2(2):69–85.
- [27] Yaiche MR, Bouhanik A. Atlas solaire algérien. CDER; 2013.
- [28] Yaiche MR, Bouhanik A. Atlas GEE 2023. CDER; 2023.
- [29] Kottek M, Grieser J, Beck C, Rudolf B, Rubel F. World map of the Köppen-Geiger climate classification updated. *Meteorol Zeitsch* 2006;15:259–63.
- [30] Rubel F, Kottek M. Observed and projected climate shifts 1901–2100 depicted by world maps of the Köppen-Geiger climate classification. *Meteorol Z* 2010;19(2):135.
- [31] Agency CI. The World Factbook 2024 Algeria. 2024, <https://www.cia.gov/the-world-factbook/countries/algeria>, [Online; Accessed 11 January 2024].
- [32] Ahmim M. Current status, distribution and conservation status of Algerian bats (mammalia: Chiroptera). *J Threat Taxa* 2017;9(1):9723–33.
- [33] Benzohra I, Megateli M, Djillali K, Makhloufi S, Benouis S, Boumaaza B, Bendahmane B. Varietal diversity of date palm (Phoenix dactylifera L.) in southwestern oases of Algerian sahara: A review. *Agric Rev* 2022;43(4):475–9.
- [34] Chehma A. Le sahara en algérie, situation et d'efis. In: L'effet du Changement Climatique sur l'élevage et la gestion durable des parcours dans les zones arides et semi-arides du Maghreb. Université KASDI MERBAH - Ouargla; 2011, p. 14–21.

- [35] Messai A, Benkedda Y, Bouaichaoui S, Benzerga M. Feasibility study of parabolic trough solar power plant under Algerian climate. *Energy Procedia* 2013;42:73–82.
- [36] Exchange HD. Algeria: High Resolution Population Density Maps + Demographic Estimates. 2022, <https://data.humdata.org/dataset/highresolutionpopulationdensitymaps-dza?>, [Online; Accessed 02 January 2024].
- [37] Ali E. Geographic information system (GIS): definition, development, applications & components. India: Department of Geography, Ananda Chandra College; 2020.
- [38] Kumar M, Singh RB, Singh A, Pravesh R, Majid SI, Tiwari A. Geographic information systems in urban planning and management. Springer Nature; 2023.
- [39] Moyroud N, Portet F. Introduction to QGIS. *QGIS Generic Tools* 2018;1:1–17.
- [40] Beck HE, Zimmermann NE, McVicar TR, Vergopolan N, Berg A, Wood EF. Present and future Köppen-Geiger climate classification maps at 1-km resolution. *Sci Data* 2018;5(1):1–12.
- [41] Mittapalli G, Gorthi K, Arathian S. Development of spatial analyst toolbar in ArcGIS. *J Adv Remote Sens Gis* 2011;1(1).
- [42] BOULFANI W. Les Ambiances Thermiques d'été dans l'habitat de la période coloniale a patio (Ph.D. thesis), Université Mohamed Khider-Biskra; 2010.
- [43] Bengherqbi H. Recommandations architecturales. Tech. rep., Ministère de l'habitat algérien; 1993.
- [44] Ould-Henia A. Choix climatiques et construction: zones arides et semi arides: maison à cour de Bou-Saada. Tech. rep., EPFL; 2003.
- [45] WorldClim. WorldClim. Maps, graphs, tables, and data of the global climate. 2020, <https://www.worldclim.org>, [Online; Accessed 23 December 2023].
- [46] Haddi F, Sadeg L, Douba O, Behaz G, Kerrouche M, Sahabi S. Conception et développement de l'application AmetDBase (Algerian meteorological DATA base). *JAMA* 2020;4:37–40.
- [47] Power N. POWER Data Methodology. 2020, <https://power.larc.nasa.gov/docs/methodology/>, [Online; Accessed 12 December 2023].
- [48] Gelaro R, McCarty W, Suárez MJ, Todling R, Molod A, Takacs L, Randles CA, Darmenov A, Bosilovich MG, Reichle R, et al. The modern-era retrospective analysis for research and applications, version 2 (MERRA-2). *J Clim* 2017;30(14):5419–54.
- [49] UNESCO. Convention concerning the protection of the world cultural and natural heritage: Fourth session, Paris (France). Tech. rep., UNESCO; 1980.
- [50] UNESCO. Convention concerning the protection of the world cultural and natural heritage: Sixth session, Paris (France). Tech. rep., UNESCO; 1982.
- [51] UNESCO. Convention concerning the protection of the world cultural and natural heritage: Sixteenth session, Santa Fe (USA). Tech. rep., UNESCO; 1992.
- [52] Leask A. World heritage site designation. In: *Managing world heritage sites*. Routledge; 2006, p. 5–19.
- [53] UNESCO. World Heritage Convention : Algeria. 2005, <https://whc.unesco.org/en/statesparties/dz>, [Online; Accessed 12 November 2023].
- [54] Zhou Q. Digital elevation model and digital surface model. *Int Encycl Geogr: People Earth Environ Technol* 2017;1:1–17.
- [55] Exchange HD. Algeria - Elevation Model. 2017, <https://data.humdata.org/dataset/algeria-elevation-model?>, [Online; Accessed 02 January 2024].
- [56] Malaperdas GD, Panagiotidis VV. The aspects of aspect: Understanding land exposure and its part in geographic information systems analysis. *Energy Environ* 2018;29(6):1022–37.
- [57] Erbs D, Klein S, Duffie J. Estimation of the diffuse radiation fraction for hourly, daily and monthly-average global radiation. *Solar Energy* 1982;28(4):293–302.
- [58] Duffie JA, Beckman WA. Solar engineering of thermal processes. John Wiley & Sons; 2013.
- [59] Tiwari GN. Advance solar photovoltaic thermal energy technologies: Fundamentals, principles, design, modelling and applications. Springer Nature; 2023.
- [60] Liu B, Jordan R. Daily insolation on surfaces tilted towards equator. *ASHRAE J* (United States) 1961;10.
- [61] Berisha X, Zeqiri A, Meha D. Solar radiation—the estimation of the optimum tilt angles for south-facing surfaces in pristina. 2017, preprint.
- [62] Amente G, Muna YB. Estimation of solar panel orientation with different tilt angles at haramaya university (Ph.D. thesis), Haramaya University; 2014.
- [63] Rejhwni B. Optimum tilt and azimuth angle for raipur, India. *Int J Adv Manage Technol Eng Sci* 2018;8:154–7.
- [64] Ajao K, Ambali R, Mahmoud M. Determination of the optimal tilt angle for solar photovoltaic panel in ilorin, Nigeria. *J Eng Sci Technol Rev* 2013;6(1):87–90.
- [65] Kocer A, Yaka IF, Sardogan GT, Gungor A. Effects of tilt angle on flat-plate solar thermal collector systems. *Brit J Appl Sci Technol* 2015;9:77–85.
- [66] Rayhan A, Kinzler R. Advancing scientific computing with python's scipy library. 2023, preprint.
- [67] Nelli F. Python data analytics with Pandas, NumPy, and Matplotlib. Springer; 2018.
- [68] Kawase M, Okajima K, Uchiyama Y, et al. Evaluation of potential geographic distribution for large-scale photovoltaic system in suburbs of China. *J Renew Energy* 2013;2013.
- [69] WMO. Guide to meteorological instruments and methods of observation. World Meteorological Organization; 2008.
- [70] Soman K, Loganathan R, Ajay V. Machine learning with SVM and other kernel methods. PHI Learning Pvt. Ltd.; 2009.
- [71] James G, Witten D, Hastie T, Tibshirani R, Taylor J. Linear regression. In: *An introduction to statistical learning: with applications in python*. Springer; 2023, p. 69–134.
- [72] Mohandes MA, Rehman S. Estimation of sunshine duration in Saudi Arabia. *J Renew Sustain Energy* 2013;5(3).
- [73] Pedregosa F, Varoquaux G, Gramfort A, Michel V, Thirion B, Grisel O, Blondel M, Prettenhofer P, Weiss R, Dubourg V, et al. Scikit-learn: Machine learning in python. *J Mach Learn Res* 2011;12:2825–30.
- [74] Rekik S, El Alimi S. GIS-AHP based approach for largescale solar PV site selection: A case of Tunisia. 2021, Available at SSRN 3977563.
- [75] Rekik S, El Alimi S. Optimal wind-solar site selection using a GIS-AHP based approach: A case of Tunisia. *Energy Convers Manage* 2023;18:100355.
- [76] Rekik S, El Alimi S. Land suitability mapping for large-scale solar PV farms in tunisia using GIS-based MCDM approach. In: *2023 IEEE international conference on artificial intelligence & green energy. ICAIGE, IEEE; 2023*, p. 1–5.
- [77] Ouchani F-z, Jbaili O, Maaroufi M, Ghennioui A. Identification of suitable sites for large-scale photovoltaic installations through a geographic information system and analytical hierarchy process combination: a case study in marrakesh-safi region, Morocco. *Prog Photovolt, Res Appl* 2021;29(7):714–24.
- [78] Jbaili O, Ouchani F-z, Merrouni AA, Cherkaoui M, Ghennioui A, Maaroufi M. An AHP-GIS based site suitability analysis for integrating large-scale hybrid CSP+ PV plants in Morocco: An approach to address the intermittency of solar energy. *J Clean Prod* 2022;369:133250.
- [79] Elboshy B, Alwetaishi M, Aly RM, Zalhaf AS. A suitability mapping for the PV solar farms in Egypt based on GIS-AHP to optimize multi-criteria feasibility. *Ain Shams Eng J* 2022;13(3):101618.
- [80] Rediske G, Siluk JCM, Michels L, Rigo PD, Rosa CB, Cugler G. Multi-criteria decision-making model for assessment of large photovoltaic farms in Brazil. *Energy* 2020;197:117167.
- [81] Khemiri W, Yaagoubi R, Miky Y. Optimal placement of solar photovoltaic farms using analytical hierarchical process and geographic information system in Mekkah, Saudi Arabia. In: *AIP conference proceedings*. Vol. 2056, AIP Publishing; 2018, 020025.
- [82] Albraheem L, Alabdulkarim L. Geospatial analysis of solar energy in riyadh using a GIS-AHP-based technique. *ISPRS Int J Geo-Inf* 2021;10(5):291.
- [83] Yushchenko A, de Bono A, Chatenoux B, Patel MK, Ray N. GIS-based assessment of photovoltaic (PV) and concentrated solar power (CSP) generation potential in West Africa. *Renew Sustain Energy Rev* 2018;81:2088–103.
- [84] Tunc A, Tuncay G, Alacakanat Z, Sevimli F. GIS based solar power plants site selection using analytic hierarchy process (ahp) in Istanbul, Turkey. *Int Arch Photogramm Remote Sens Spat Inf Sci* 2019;42:1353–60.
- [85] Colak HE, Memisoglu T, Gercek Y. Optimal site selection for solar photovoltaic (PV) power plants using GIS and AHP: A case study of Malatya Province, Turkey. *Renew Energy* 2020;149:565–76.
- [86] Günen MA. A comprehensive framework based on GIS-AHP for the installation of solar PV farms in Kahramanmaraş, Turkey. *Renew Energy* 2021;178:212–25.
- [87] Marques-Perez I, Guaita-Pradas I, Gallego A, Segura B. Territorial planning for photovoltaic power plants using an outranking approach and GIS. *J Clean Prod* 2020;257:120602.
- [88] López-Bravo C, Mora-López L, Sidrach-deCardona M, Márquez-Ballesteros MJ. A comprehensive analysis based on GIS-AHP to minimise the social and environmental impact of the installation of large-scale photovoltaic plants in south Spain. *Renew Energy* 2024;226:120387.
- [89] Rios R, Duarte S. Selection of ideal sites for the development of large-scale solar photovoltaic projects through Analytical Hierarchical Process–Geographic information systems (AHP-GIS) in Peru. *Renew Sustain Energy Rev* 2021;149:111310.
- [90] Noorollahi Y, Senani AG, Fadaei A, Simaee M, Moltames R. A framework for GIS-based site selection and technical potential evaluation of PV solar farm using fuzzy-boolean logic and AHP multi-criteria decision-making approach. *Renew Energy* 2022;186:89–104.
- [91] Ruiz H, Sunarso A, Ibrahim-Bathis K, Murti S, Budiarto I. GIS-AHP multi criteria decision analysis for the optimal location of solar energy plants at Indonesia. *Energy Rep* 2020;6:3249–63.
- [92] Anselin L, Syabri I, Kho Y. Geoda: an introduction to spatial data analysis. In: *Handbook of applied spatial analysis: software tools, methods and applications*. Springer; 2009, p. 73–89.
- [93] Arulloss M, Lakshmi TM, Venkatesan VP. A survey on multi criteria decision making methods and its applications. *Am J Inf Syst* 2013;1(1):31–43.
- [94] Thakkar JJ. Multi-criteria decision making. Vol. 336, Springer; 2021.
- [95] Canco I, Kruija D, Iancu T. AHP, a reliable method for quality decision making: A case study in business. *Sustainability* 2021;13(24):13932.
- [96] Hruska R, Prusa P, Babic D. The use of AHP method for selection of supplier. *Transport* 2014;29(2):195–203.
- [97] Dos Santos M, de Araújo Costa IP, Gomes CFS. Multicriteria decision-making in the selection of warships: a new approach to the AHP method. *Int J Anal Hierarchy Process* 2021;13(1).

- [98] Al-Azab FGM, Ayu MA. Web based multi criteria decision making using AHP method. In: Proceeding of the 3rd international conference on information and communication technology for the moslem world (ICT4M) 2010. IEEE; 2010, p. A6–12.
- [99] Petrillo A, De Felice F. Analytic Hierarchy Process-Models, Methods, Concepts, and Applications: Models, Methods, Concepts, and Applications. BoD-Books on Demand; 2023.
- [100] Saaty TL, Vargas LG, Saaty TL, Vargas LG. The seven pillars of the analytic hierarchy process. In: Models, methods, concepts & applications of the analytic hierarchy process. Springer; 2012, p. 23–40.
- [101] Nijkamp P, Beinat E. Multicriteria analysis for land-use management. Kluwer Academic Publishers; 1998.
- [102] Malczewski J. On the use of weighted linear combination method in GIS: common and best practice approaches. *Trans GIS* 2000;4(1):5–22.
- [103] Dixon B, Uddameri V. GIS and geocomputation for water resource science and engineering. John Wiley & Sons; 2016.
- [104] Blachowski J, Rybakiewicz W, Warczewski W, Malczewski P. Application of multi-criteria analysis in GIS for optimal planning of house development areas. Case study of wrocław functional area. *Roczniki Geomatyki* 2016;14(5 (75):561–71.
- [105] PEAKLIST. A website for the mountains of the world based on prominence. 2004, <http://peaklist.org/WWlists/ultras/africa.html>, [Online; Accessed 12 September 2024].
- [106] Scotto E. The longest ride: my ten-year 500,000 mile motorcycle journey. 2013, (No Title).
- [107] pouretudiant. Les montagnes d'Atlas. 2012, <https://pouretudiant.blogspot.com/2012/12/les-montagnes-datlas.html>, [Online; Accessed 27 August 2024].
- [108] Hughes RH. A directory of African wetlands. IUCN; 1992.
- [109] Larousse. Kabylie des Babors. 2024, <https://www.larousse.fr/encyclopedie/mont/Babors/107041>, [Online; Accessed 27 August 2024].
- [110] Noor-book. Mont isaak. 2024, <https://www.noor-book.com/tag/>, [Online; Accessed 27 August 2024].
- [111] Tchoumatchenco P. Brachiopodes du Jurassique inferieur et moyen du Kef Sidi Amar–Massif culminant de l'Ouarsenis (Algerie du Nord). *Geol Balc* 1994;24(1):25–61.
- [112] Bourouba M. Les variations de la turbidité et leurs relations avec le débit et les précipitations dans les bassins-versants des oueds reboa et boudouaou (algérie). *Bologne (Italie)* 1997;52–8.
- [113] Petitfute. Pic des singes. 2019, <https://www.petitfute.com/v62981-parc-national-de-gouraya/c1173-visites-points-d-interet/c974-site-naturel/109815-pic-des-singes.html>, [Online; Accessed 27 August 2024].
- [114] Naimi B. Etat des lieux de la Biodiversite de Djebel Ksel dans la region d'El-Bayadh (Ph.D. thesis), Tlemcen University; 2020.
- [115] Wikipedia. Titteri. 2023, <https://en.wikipedia.org/wiki/Titteri>, [Online; Accessed 27 August 2024].
- [116] Britannica. Chott melrhir. 1998, <https://www.britannica.com/place/Chott-Melrhir>, [Online; Accessed 27 August 2024].
- [117] Abdelouahab B, Belaid M, Cherchali M, Moulla A, Said R, Chorfi H. Caractérisation hydrogéochimique et isotopique de la nappe calcaire du zaccar. In: Séminaire international sur l'hydrogéologie et l'environnement. Université Ouargla; 2019, p. 38–42.
- [118] D'Outre-Mer AN. Communes mixtes d'Algérie. Commune mixte des Bibans. 2016, <https://recherche-anom.culture.gouv.fr/ark:/61561/aw655jddfe>, [Online; Accessed 27 August 2024].
- [119] Alayat H, El Khattabi J, Lamouroux C. Evolution spatiale des caracteristiques physico-chimiques des eaux du lac oubeira impose par les conditions severes de la secheresse (extreme ne algerien). *Eur Sci J* 2013;9(36).
- [120] éditorial du programme STRABON E. Chott el hodna. 1998, <https://shorturl.at/ckXO2>, [Online; Accessed 27 August 2024].
- [121] Amri C, Neffar S, Ouchtati N, Chenchouni H. Spatiotemporal patterns of ground beetle diversity (coleoptera: Carabidae) in a ramsar wetland (chott tinsilt) of Algeria. *Turk J Zool* 2019;43(5):502–15.
- [122] Kim S, Kim H. A new metric of absolute percentage error for intermittent demand forecasts. *Int J Forecast* 2016;32(3):669–79.
- [123] Condor. Solar kits catalog. SPA Condor Electronics; 2018.